Provably Scalable Black-Box Variational Inference with Structured Variational Families

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Abstract

Variational families with full-rank covariance approximations are known not to work well in black-box variational inference (BBVI), both empirically and theoretically. In fact, recent computational complexity results for BBVI have established that full-rank variational families scale poorly with the dimensionality of the problem compared e.g. to mean field families. This is particularly critical to Hierarchical Bayesian models with local variables: their dimensionality increases with the size of the datasets. In this paper, we explore the theoretical middle ground *between* mean-field variational families and full rank families. In particular, we explored *structured* covariance approximations that are more expressive than mean-field but that still enjoy fast empirical and theoretical convergence properties. We rigorously prove that certain structured covariances can achieve a computational complexity independent of the dataset size, implying scalability. We empirically verify our theoretical results on real-world Hierarchical Bayesian models.

1 INTRODUCTION

A decade has passed since black-box variational inference (BBVI; Ranganath et al., 2014; Titsias and Lázaro-Gredilla, 2014), also known as Monte Carlo variational inference and stochastic gradient variational Bayes, has emerged. Among various approaches to variational inference (VI; Blei et al., 2017; Jordan et al., 1999; Zhang et al., 2019), BBVI has been immensely successful in various fields such as statistics, machine learning, signal processing, and many more, thanks to its general black-box nature and scalability to large datasets.

One of the promises of BBVI has been that we can better model correlations in the posterior by using full-rank covariance approximations (Kucukelbir et al., 2017). That way, we should have been able to move away from mean-field (Hinton and van Camp, 1993; Peterson and Anderson, 1987), or diagonal covariance, approximations traditionally required when performing coordinate-ascent VI (CAVI; see the review by Blei et al. 2017). However, this promise has shown to be elusive. Despite their theoretical and occasional empirical superiority in terms of "expressiveness," going full-rank does not always improve over mean-field. See, for example, the experiments in §3 by Zhang et al. (2022), §B.1 by Agrawal et al. (2020), Footnote 2 by Giordano et al. (2018).

A common explanation for the underwhelming performance of full-rank approximations has been their excessive gradient variance. Indeed, recent results by Domke et al. (2023); Kim et al. (2023b) establishing the computational complexity of BBVI confirm this intuition. Due to their gradient variance, full-rank covariance approximations result in a $\mathcal{O}\left(d\kappa^2\epsilon^{-1}\right)$ iteration complexity for finding an ϵ-accurate solution on strongly log-concave posteriors with a condition number of κ. This contrasts with mean-field for which a $\mathcal{O}\left(\sqrt{d}\right)$ dimensional dependence (Kim et al., 2023c) have been established. The poor dimension dependence explains the poor behavior of full-rank covariance approximations in practice.

Furthermore, for models with *local variables* (Hoffman and Blei, 2015), the $\mathcal{O}(d)$ dimension dependence is especially concerning; for these models the dimensionality d scales with the size of the dataset N. This means that, for a model taking $\mathcal{O}(N)$ time to evaluate the joint likelihood, the sample complexity of BBVI will be $\mathcal{O}\left(N^2\epsilon^{-1}\right)$. In terms of the scalability with respect to the size of dataset, this highlights a clear challenge for BBVI with full-rank variational families.

While mean-field is scalable, it is a crude approximation as it fails to model correlations in the posterior. Therefore, a natural question is, "Can we find a variational family more expressive than mean-field families while maintaining computational tractability?" In this paper, based on the theoretical framework of Domke et al. (2023); Kim et al.

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(2023b), we answer this question by revisiting the idea of *structured covariances*.

Structured covariance variational families, or more broadly, structured variational families, exploit structural assumptions about the factor graph of the posterior, and have been a widely explored concept in VI (Hoffman and Blei, 2015; Ranganath et al., 2016; Saul and Jordan, 1995; Tan and Nott, 2018; for a short representative list; see Section 5 for an extended list). However, why and when structured covariances can outperform full-rank ones has not been rigorously analyzed. For instance, Tan and Nott, who first performed a detailed investigation into structured covariances in BBVI, only mentioned that "unrestricted (full-rank) Gaussian variational approximation . . . can be prohibitively slow for large data since the number of variational parameters scales as the square of the length of $\boldsymbol{z}$." However, given the incredible success of stochastic optimization in the over-parameterized regime, simply having fewer parameters does not fully explain why structured covariance families work well; a deeper understanding is desirable.

Based on the framework of Domke et al. (2023); Kim et al. (2023b), we rigorously prove that structured covariance variational families can improve the dimensional dependence of full-rank variational families. For Hierarchical Bayesian models with local variables, we provide a covariance structure that can entirely remove the dependence on the dataset size N. In more detail, for the canonical 1-level hierarchy where the variables can be split into global and local variables, we show that a Cholesky-factor parameterization with a bordered block-diagonal structure completely resolves the dependency on N. This extends the current line of work of proving that the choice of the variational family is solving a trade-off between statistical accuracy and computational efficiency (Bhatia et al., 2022; Kim et al., 2023c).

❶ Theorem 1 establishes that, for location-scale variational families, manipulating the covariance structure can improve the dimensional dependence of BBVI.

❷ From Theorem 1, for Hierarchical Bayesian models with local variables, Corollary 2 establishes that the covariance structure previously proposed by Tan (2021); Tan et al. (2020) achieves a computational complexity independent of the dataset size.

2 PRELIMINARIES

Notation Random variables are denoted in serif, vectors in bold, and matrices in bold capitals. (*i.e.,* x, $\boldsymbol{x}$, and $\boldsymbol{A}$ respectively.) Given a vector $\boldsymbol{x} \in \mathbb{R}^d$, we denote its Euclidean norm as $\|\boldsymbol{x}\|_2 = \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle} = \sqrt{\boldsymbol{x}^\top \boldsymbol{x}}$. For a matrix $\boldsymbol{A}$, we denote its Frobenius norm as $\|\boldsymbol{A}\|_{\mathrm{F}} = \sqrt{\mathrm{tr}\left(\boldsymbol{A}^\top \boldsymbol{A}\right)}$, where $\mathrm{tr}\left(\boldsymbol{A}\right) = \sum_{i=1}^{d} \boldsymbol{A}_{ii}$, and the ℓ_2-operator norm as $\|\boldsymbol{A}\|_{2,2}$. Also, for some $\boldsymbol{B} \succ 0$, $\|\boldsymbol{x}\|_{\boldsymbol{B}} = \sqrt{\boldsymbol{x}^\top \boldsymbol{B} \boldsymbol{x}}$, and $\boldsymbol{A} \succ \boldsymbol{B}$ is the Loewner order such that $\boldsymbol{A} - \boldsymbol{B}$ is positive-definite. Lastly, $\mathrm{GL}_d\left(\mathbb{R}\right)$ denotes the set of d-dimensional triangular matrices.

2.1 Variational Inference

Variational inference (VI; Blei et al., 2017; Jordan et al., 1999; Zhang et al., 2019) is a method for approximating an intractable probability distribution through optimization. In general, we aim to minimize the Kullback-Leibler (KL) divergence as described by the following:

$$\underset{\boldsymbol{\lambda} \in \Lambda}{\text{minimize}} \;\; \mathrm{D}_{\mathrm{KL}}\left(q_{\boldsymbol{\lambda}}, \pi\right),$$

where	D_{KL}	is the KL divergence,
	π	is the (target) posterior distribution, and
	$q_{\boldsymbol{\lambda}}$	is the variational approximation.

In the context of Bayesian inference, we only have access to the joint likelihood $p\left(\boldsymbol{z}, \boldsymbol{x}\right) \propto \pi\left(\boldsymbol{z}\right)$. This results in the KL divergence also being intractable due to. Therefore, we instead rely on minimizing a surrogate objective called the negative *evidence lower bound* (ELBO, Jordan et al., 1999) function $F(\boldsymbol{\lambda})$:

$$\underset{\boldsymbol{\lambda} \in \Lambda}{\text{minimize}} \;\; F\left(\boldsymbol{\lambda}\right) \triangleq -\mathbb{E}_{\mathsf{z} \sim q_{\boldsymbol{\lambda}}} \log p\left(\mathsf{z}, \boldsymbol{x}\right) - \mathbb{H}\left(q_{\boldsymbol{\lambda}}\right),$$

where	$p\left(\mathsf{z}, \boldsymbol{x}\right)$	is the joint likelihood,
	$\mathbb{H}$	is the differential entropy.

Without loss of generality, we will also separately denote the negative log joint likelihood as

$$\ell\left(\mathsf{z}\right) \triangleq -\log p\left(\mathsf{z}, \boldsymbol{x}\right).$$

Under this notation, the ELBO can be represented as a *composite* optimization problem. (See the taxonomization in §2.4 by Kim et al. 2023b.)

$$F\left(\boldsymbol{\lambda}\right) = f\left(\boldsymbol{\lambda}\right) + h\left(\boldsymbol{\lambda}\right),$$

where $f(\boldsymbol{\lambda}) \triangleq \mathbb{E}_{\mathsf{z} \sim q_{\boldsymbol{\lambda}}} \ell\left(\mathsf{z}\right)$, is called the *energy* and $h\left(\boldsymbol{\lambda}\right) \triangleq -\mathbb{H}\left(q_{\boldsymbol{\lambda}}\right)$ is an entropic regularizer.

2.2 Variational Family

We focus on the location-scale variational family:

Definition 1 (Location-Scale Family). Let φ be some d-variate distribution. Then, $q_{\boldsymbol{\lambda}}$ that can be equivalently represented as

$$\mathsf{z} \sim q_{\boldsymbol{\lambda}} \quad \Leftrightarrow \quad \mathsf{z} \overset{d}{=} \mathcal{T}_{\boldsymbol{\lambda}}\left(\mathsf{u}\right); \quad \mathsf{u} \sim \varphi,$$

where $\overset{d}{=}$ is equivalence in distribution, is said to be part of the location-scale family generated by the base distribution φ and the reparameterization function $\mathcal{T} : \Lambda \times \mathbb{R}^d \to \mathbb{R}^d$ defined as

$$\mathcal{T}_{\boldsymbol{\lambda}}\left(\mathsf{u}\right) \triangleq \boldsymbol{C}\mathsf{u} + \boldsymbol{m}$$

with $\boldsymbol{\lambda} \in \Lambda \subseteq \mathbb{R}^p$ containing the parameters for forming the location $\boldsymbol{m} \in \mathbb{R}^d$ and scale $\boldsymbol{C} \in \mathbb{R}^{d \times d}$.

Since the reparameterization function is differentiable, this enables the use of the *reparameterization gradient* (Kingma

and Welling, 2014; Titsias and Lázaro-Gredilla, 2014). We define the M-sample reparameterization gradient estimator $\boldsymbol{g}(\boldsymbol{\lambda})$ to be an unbiased estimator of the gradient of the energy $\nabla f(\boldsymbol{\lambda})$ as:

$$\boldsymbol{g}_M(\boldsymbol{\lambda}) \triangleq \frac{1}{M} \sum_{m=1}^{M} \nabla_{\boldsymbol{\lambda}} \ell(\mathcal{T}_{\boldsymbol{\lambda}}(\boldsymbol{u}_m)), \tag{1}$$

where $\boldsymbol{u}_m \sim \varphi$ is a Monte Carlo sample. The reparameterization gradient is empirically known to result in lower variance compared to other gradient estimators (Mohamed et al., 2020; Xu et al., 2019) and is presently *de-facto* standard for BBVI.

For the base distribution φ, the choice of distribution generates various other families (Titsias and Lázaro-Gredilla, 2014) used in practice: Mean-field Gaussian, full-rank Gaussian, Laplace, Student-t, and many more. For our theoretical analysis, we impose mild assumptions on the base distribution φ:

Assumption 1 (Base Distribution). φ is a d-dimensional distribution such that $\boldsymbol{u} \sim \varphi$ and $\boldsymbol{u} = (u_1, \dots, u_d)$ with indepedently and identically distributed components. Furthermore, φ is **(i)** symmetric and standardized such that $\mathbb{E}u_i = 0$, $\mathbb{E}u_i^2 = 1$, $\mathbb{E}u_i^3 = 0$, and **(ii)** has finite kurtosis $\mathbb{E}u_i^4 = k_\varphi < \infty$.

These assumptions are satisfied by most practically used location-scale families.

In summary, the assumptions related to the variational family are enumerated below:

Assumption 2. The variational family is the location-scale family formed by Definition 1 with the base distribution φ satisfying Assumption 1.

2.3 Scale Parameterization

The parameterization of the scale matrix in location-scale families $\boldsymbol{C}$ can result in vastly different statistical and computational performance results. In principle, the only restriction is that it needs to result in a proper covariance matrix such that $\boldsymbol{C}\boldsymbol{C}^\top \succ 0$. However, how many low-rank factors we include into $\boldsymbol{C}$ corresponds to a statistical-computational trade-off (Bhatia et al., 2022; Ong et al., 2018). A common choice is to restrict $\boldsymbol{C}$ to be a diagonal matrix, resulting in the *mean-field* approximation (Hinton and van Camp, 1993; Peterson and Anderson, 1987).

Triangular Scale While the mean-field approximation results in fast convergence (Kim et al., 2023c) and stable optimization, it is a crude approximation as it ignores correlations in the posterior. Therefore, we are interested in scale matrices that are more complex than mean-field. For this, we will first restrict our interest to (lower) triangular scale matrices such that $\boldsymbol{C} \in \mathrm{GL}_d(\mathbb{R})$. Other choices allowing theoretical analysis include the "matrix square-root" parameterization used by (Domke et al., 2023). But the triangular, or Cholesky, parameterization has additional benefits:

(a) It results in lower gradient variance than the matrix square-root parameterization (Kim et al., 2023c),
(b) computing the entropy term h can be done in $\mathcal{O}(d)$ time,
(c) the positive definiteness of $\boldsymbol{C}$ can be enforced only by manipulating the d diagonal elements, and
(d) The conditional dependence between the coordinates can be manipulated through the matrix elements.

Naturally, Item (d) is important for setting structure covariances.

Dealing with Non-Smoothness of the Entropy h The triangular parameterization also has downsides. While in practice, (projected) SGD is often used for BBVI, there is currently no known way to prove its convergence triangular matrices. This is because a projection operator that maps to the space of triangular matrices while ensuring h is smooth is currently unknown. One could instead use the matrix square-root parameterization, but then, the complexity of performing a projection is $\mathcal{O}(d)$. For this reason, to stick with the Cholesky parameterization, we will instead rely on proximal methods as explained in Section 2.4.

2.4 Proximal Stochastic Gradient Descent

In this work, we optimize $F(\boldsymbol{\lambda})$ using stochastic gradient descent (SGD; Bottou, 1999; Nemirovski et al., 2009; Robbins and Monro, 1951). For reasons discussed in Section 2.3, we will focus on proximal SGD (Ghadimi and Lan, 2012).

Proximal SGD Proximal SGD, or more commonly stochastic proximal gradient descent, aims to minimize a composite objective expressed as the sum, $f + h$, where f is smooth. Given initial parameters $\boldsymbol{\lambda}_0$ and a step size schedule $(\gamma_t)_{t=0}^{T-1}$, proximal SGD is repeats the update:

$$\boldsymbol{\lambda}_{t+1} = \mathrm{prox}_{\gamma_t, h}(\boldsymbol{\lambda}_t - \gamma_t \boldsymbol{g}(\boldsymbol{\lambda}_t))$$

until convergence, where $\boldsymbol{g}(\boldsymbol{\lambda}_t)$ is a stochastic estimate of ∇f, and prox is a *proximal operator* defined as:

$$\mathrm{prox}_{\gamma_t, h} \triangleq \arg\min_{\boldsymbol{v}} \left[h(\boldsymbol{v}) + \frac{1}{2\gamma_t} \|\boldsymbol{\lambda} - \boldsymbol{v}\|_2^2 \right].$$

Proximal SGD in BBVI Proximal SGD has been proposed for variational inference under different motivations (Altosaar et al., 2018; Diao et al., 2023; Khan et al., 2016, 2015). However, Domke (2019) first suggested proximal SGD to circumvent the fact that the ELBO is non-smooth due to the entropic regularizer h being non-smooth. While the non-smoothness of the ELBO can be solved through alternative means, *e.g.*, projected SGD, nonlinear parameterizations, projected SGD necessitate the choice of a restricted domain, and nonlinear parameterizations result in slower convergence rate (Kim et al., 2023b). As such, we will focus on proximal SGD.

3 THEORETICAL ANALYSIS

3.1 Fundamental Limits of Being Full-Rank

First, we will discuss what we currently know about the dimensional dependence of BBVI. We will also illustrate exactly *where* the dimensional dependence is coming from. This should make the solution more clear.

Gradient Variance For an L-log-smooth and μ-strongly log-concave target posterior π and a full-rank variational family, that is, $\boldsymbol{C}$ is a dense or triangular matrix, Domke (2019, Eq. 3, Thm. 3) show that the stochastic gradient estimator with $M = 1$ satisfies

$$\mathbb{E}\|\boldsymbol{g}_1(\boldsymbol{\lambda})\|_2^2 \leq L^2 \left(d + k_\varphi\right) \|\boldsymbol{\lambda} - \bar{\boldsymbol{\lambda}}\|_2^2, \tag{2}$$

for some $\bar{\boldsymbol{\lambda}} \in \mathbb{R}^p$.

Iteration Complexity Using Equation (2), Domke et al. (2023) recently establish the following:

Theorem (Domke et al. 2023)**.** *The iteration complexity of finding an ϵ-accurate solution $\boldsymbol{\lambda}_T$ such that $\|\boldsymbol{\lambda}_T - \boldsymbol{\lambda}^*\|_2^2 \leq \epsilon$, where $\boldsymbol{\lambda}^*$ is the global minimizer, with proximal SGD BBVI and a fixed stepsize is*

$$\mathcal{O}\left(\kappa^2 d \frac{1}{\epsilon} \log\left(\Delta_0^2 \frac{1}{\epsilon}\right)\right),$$

where $\kappa = L/\mu$ and $\Delta_0 = \|\boldsymbol{\lambda}_0 - \boldsymbol{\lambda}^*\|_2$ is the distance between the initial point $\boldsymbol{\lambda}_0$ and $\boldsymbol{\lambda}^*$.

Local Variables In probabilistic modeling, some Hierarchical models have latent variables, $\boldsymbol{y}_n$ for $n = 1, \ldots, N$, unique to each datapoint $\boldsymbol{x}_n$ (Hoffman and Blei, 2015). For example, in mixture and mixed-membership models, each datapoint has a latent variable indicating from which mixture component it was generated. In state space models, at each time point t, the state is often not observed directly and is a latent variable local to t.

More abstractly, consider a model with local variables $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_N$ and a global variable $\boldsymbol{z}$ such that $\boldsymbol{y}_n \in \mathbb{R}^{d_y}$ and $\boldsymbol{z} \in \mathbb{R}^{d_z}$. The total dimension of the model is

$$d = N d_y + d_z.$$

Then, for these models, the complexity ends up with a dependence on the size of the dataset.

Corollary 1 (informal)**.** *For a Hierarchical Bayesian model with $\mathcal{O}(N)$ local variables, where N is the number of datapoints in the dataset, we have $d = \mathcal{O}(N)$. Therefore, the complexity is*

$$\mathcal{O}\left(N \kappa^2 \frac{1}{\epsilon} \log\left(\Delta_0^2 \frac{1}{\epsilon}\right)\right).$$

Remark 1. For models where the complexity of evaluating the gradient of the joint likelihood is $\mathcal{O}(N)$, the total complexity of BBVI with proximal SGD is $\mathcal{O}\left(N^2\right)$.

Clearly, full-rank families do not scale even if we ignore the $\mathcal{O}\left(N^2\right)$ storage requirement of the covariance.

Remark 2 (**Are less parameters *obviously* better?**)**.** The number of variational parameters p enters the complexity statement through Δ_0. Therefore, the dependence on the number of parameters alone is only logarithmic. This implies that simply reducing the number of parameters does not obviously improve the dimensional dependence.

Remark 3 (**Where does d come from?**)**.** The $\mathcal{O}(d)$ dimension dependence directly comes from Equation (2), the gradient variance. Therefore, reducing the gradient variance is key to improving the complexity of BBVI.

3.2 Complexity of BBVI with Block Structured Scale Matrices

Recall that $\boldsymbol{z}$ is a d-dimensional vector that denotes the model parameters. We consider a partitioning of $\boldsymbol{z}$ into B blocks of varying sizes such that

$$\boldsymbol{z} = (\boldsymbol{z}_1, \ldots, \boldsymbol{z}_b, \ldots, \boldsymbol{z}_B),$$

where $\boldsymbol{z}_b \in \mathbb{R}^{d_{z_b}}$ such that $\sum_{b=1}^B d_{z_b} = d$. Such partitioning is often naturally possible depending on the structure of the model ℓ. We will focus on a special case later.

Block-Structured Scale Matrix The location-scale reparameterization function $\mathcal{T}$ can be block partitioned over blocks of rows to match this structure. That is,

$$\begin{bmatrix} \boldsymbol{z}_1 \\ \vdots \\ \boldsymbol{z}_b \\ \vdots \\ \boldsymbol{z}_B \end{bmatrix} = \begin{bmatrix} \boldsymbol{C}_{z_1,z_1} & & & & \\ \vdots & \ddots & & & \\ \boldsymbol{C}_{z_b,z_1} & \cdots & \boldsymbol{C}_{z_b,z_b} & & \\ \vdots & \ddots & \vdots & \ddots & \\ \boldsymbol{C}_{z_B,z_1} & \cdots & \boldsymbol{C}_{z_b,z_B} & \cdots & \boldsymbol{C}_{z_B,z_B} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_{z_1} \\ \vdots \\ \boldsymbol{u}_{z_b} \\ \vdots \\ \boldsymbol{u}_{z_B} \end{bmatrix} + \begin{bmatrix} \boldsymbol{m}_{z_1} \\ \vdots \\ \boldsymbol{m}_{z_b} \\ \vdots \\ \boldsymbol{m}_{z_B} \end{bmatrix}$$

where the grey area demarks the non-zero entries of $\boldsymbol{C}$, for $\boldsymbol{u}_b \in \mathbb{R}^{d_{z_b}}$, $\boldsymbol{m}_b \in \mathbb{R}^{d_{z_b}}$, $\boldsymbol{C}_{z_b,z_{b'}} \in \mathbb{R}^{d_{z_b} \times d_{z_{b'}}}$, and the diagonal blocks $\boldsymbol{C}_{z_b,z_b}$ are lower-triangular such that $\boldsymbol{C}_{z_b,z_b} \in \mathrm{GL}_{d_{z_b}}(\mathbb{R})$.

Furthermore, for clarity, we denote each row-wise block corresponding to $\boldsymbol{z}_b$ for some b as

$$\boldsymbol{C}_{z_b} \triangleq \left[\boldsymbol{C}_{z_b,z_1} \quad \ldots \quad \boldsymbol{C}_{z_b,z_b} \quad \boldsymbol{0} \quad \ldots \quad \boldsymbol{0}\right],$$

where $\boldsymbol{C}_{z_b} \in \mathbb{R}^{d_{z_b} \times d}$ and $\boldsymbol{0}$ is a zero block. The key quantity that characterizes the effect of each block on the gradient variance is

$$r_{z_b,z_{b'}} \triangleq \mathrm{nnzc}\left(\boldsymbol{C}_{z_b,z_{b'}}\right), \tag{3}$$

the number of non-zero columns of the block $\boldsymbol{C}_{z_b,z_{b'}}$.

Upper Bound on the Gradient Variance We will see how this block structure affects the gradient variance.

Theorem 1. *Let Assumption 2 hold. Then, for an L-smooth ℓ, the gradient variance of $\boldsymbol{g}_M(\lambda)$ is bounded*

as

$$\mathbb{E}\|\boldsymbol{g}_M(\boldsymbol{\lambda})\|_2^2 \leq \frac{L^2}{M}\left(2\left(d^* + k_\varphi\right) + M\right)\|\boldsymbol{\lambda} - \boldsymbol{\lambda}^*\|_2^2 + \frac{2L^2}{M}\left(d^* + k_\varphi\right)\|\boldsymbol{\lambda}^* - \bar{\boldsymbol{\lambda}}\|_2^2$$

for any $\boldsymbol{\lambda}, \boldsymbol{\lambda}^* \in \Lambda$, *where*

$$d^* = \max_{b=1,\dots,B} \sum_{b'=1}^{B} r_{z_b, z_{b'}}$$

is the effective dimensionality, $\bar{\boldsymbol{\lambda}} = (\bar{\boldsymbol{z}}, \boldsymbol{0})$, *and* $\bar{\boldsymbol{z}}$ *is a stationary point of* ℓ.

Proof. See the *full proof* in page 24.

Remark 4. When $\boldsymbol{C}$ is densely full-rank, such as in full-rank variational families, we have $d^* = d$. Therefore, we retrieve the dimension dependence of Domke (2019) in Equation (2). This shows that Theorem 1 generalizes existing results.

Remark 5. The intuition for understanding $r_{z_b, z_{b'}}$ is as follows: by setting a block $\boldsymbol{C}_{z_b, z_{b'}}$ to be *non-zero*, we are effectively deciding to use $\boldsymbol{u}_{b'}$ when computing $\boldsymbol{z}_b$. The exact amount of randomness that $\boldsymbol{C}_{z_b}$ mixes into $\boldsymbol{z}_b$ directly translates into dimension dependence.

Remark 6. Theorem 1 answers Remark 2. That is, the structure of $\boldsymbol{C}$, not the number of parameters in $\boldsymbol{C}$, determines the effective dimension dependence.

Notice that Theorem 1 is pointing towards improving the dimensional dependence. We can shape the resulting dimension dependence by designing the structure of $\boldsymbol{C}$. We will later show a specific structure for models with local variables that can resolve the dependence on the number of datapoints.

Complexity of Proximal SGD To obtain a convergence rate from Theorem 1, we refine the results of Domke et al. (2023) which is based on the following condition:

Assumption 3 (**Quadratic Variance**). A gradient estimator $\boldsymbol{g}(\boldsymbol{\lambda})$ of f is said to satisfy the quadratic variance (QV) condition if its expected-squared norm is bounded as

$$\mathbb{E}\|\boldsymbol{g}(\boldsymbol{\lambda})\|_2^2 \leq \alpha\|\boldsymbol{\lambda} - \bar{\boldsymbol{\lambda}}\|_2^2 + \beta$$

for any $\boldsymbol{\lambda} \in \Lambda$ and some constants $0 \leq \alpha, \beta < \infty$, where $\boldsymbol{\lambda}^* \in \Lambda$ is a stationary point of f.

It is clear that Theorem 1 satisfies Assumption 3.

We establish the precise computational complexity of proximal SGD with computable constants.

Lemma 1. *For a* μ*-strongly convex and* L*-smooth* $f : \Lambda \to \mathbb{R}$ *on a convex set* Λ *and an estimator satisfying Assumption 3, the last iterate* $\boldsymbol{\lambda}_{T+1}$ *of proximal SGD with a fixed stepsize satisfies* $\|\boldsymbol{\lambda}_{T+1} - \boldsymbol{\lambda}^*\|_2^2 \leq \epsilon$ *if*

$$\gamma = \min\left(\frac{\epsilon\mu}{4\left(\beta + L^2\bar{\Delta}^2\right)}, \frac{2\alpha}{\mu}, \mu\right) \quad \textit{and}$$

$$T \geq \max\left(\frac{4\left(\beta + L^2\bar{\Delta}^2\right)}{\epsilon\mu^2}, \frac{2\alpha}{\mu^2}, 1\right)\log\left(2\Delta_0^2\frac{1}{\epsilon}\right),$$

where $\bar{\boldsymbol{\lambda}}$ *is a stationary point of* f, $\bar{\Delta} = \|\boldsymbol{\lambda}^* - \bar{\boldsymbol{\lambda}}\|_2$, *and* $\Delta_0 = \|\boldsymbol{\lambda}^* - \boldsymbol{\lambda}_0\|_2$.

Proof. See the *full proof* in page 26.

Combined with Theorem 1, we now obtain a general complexity guarantee for BBVI:

Theorem 2. *Let Assumption 2 hold. Then, for a* μ*-strongly log-concave and* L*-log-smooth posterior, the last iterate* $\boldsymbol{\lambda}_{T+1}$ *of BBVI with proximal SGD and an* M*-sample gradient estimator* M *satisfies a suboptimality of* $\|\boldsymbol{\lambda}_{T+1} - \boldsymbol{\lambda}^*\|_2^2 \leq \epsilon$ *if*

$$T \geq 8\frac{\kappa^2}{M}\left(d^* + k_\varphi + \frac{1}{2}M\right)\max\left(\bar{\Delta}^2\frac{1}{\epsilon}, \frac{1}{2}\right)\log\left(2\Delta_0^2\frac{1}{\epsilon}\right)$$

for some fixed stepsize γ, *where* d^* *is the effective dimensionality defined in Theorem 1,* $\Delta_0 = \|\boldsymbol{\lambda}_0 - \boldsymbol{\lambda}^*\|_2$ *is the distance to the optimum,* $\bar{\Delta} = \|\bar{\boldsymbol{\lambda}} - \boldsymbol{\lambda}^*\|_2$ *is the distance between the optimum and the stationary point of* f, $\kappa = L/\mu$ *is the condition number,* $\bar{\boldsymbol{\lambda}}$ *is the stationary point of the energy* f, *and* $\boldsymbol{\lambda}^*$ *is the global optimum.*

Proof. See the *full proof* in page 27.

Remark 7. Theorem 2 shows that, for μ-strongly log-concave and L-log-smooth posteriors, BBVI with proximal SGD have an iteration complexity of

$$\mathcal{O}\left(\kappa^2 d^* \frac{1}{\epsilon}\log\left(\frac{1}{\epsilon}\right)\right).$$

Remark 8. It is also possible to prove a $\mathcal{O}\left(\kappa^2 d^* \, 1/\epsilon\right)$ complexity using a decreasing stepsize schedule of Gower et al. (2019) as shown by Domke et al. (2023).

3.3 Scaling BBVI on Hierarchical Bayesian Models via Structured Covariances

Equipped with the results of Section 3.2, we present a scale matrix structure that is provably scalable in terms of computational complexity.

Consider a canonical 1-level Hierarchical model with local variables $\boldsymbol{y}_1, \dots, \boldsymbol{y}_N$ and a global variable $\boldsymbol{z}$ such that $\boldsymbol{y}_n \in \mathbb{R}^{d_y}$ and $\boldsymbol{z} \in \mathbb{R}^{d_z}$. (All local variables are assumed to have the same dimensionality.)

Structured Covariance Matrices We assume the variables are structured as

$$\begin{bmatrix}\boldsymbol{z}^\top & \boldsymbol{y}_1^\top & \dots & \boldsymbol{y}_N^\top\end{bmatrix}.$$

Table 1: Models and Datasets used in the Experiments

PROBLEM	DIMENSIONS				# OF VARIATIONAL PARAMETERS (p)		
	N	d_y	d_z	d	mean-field	structured	full-rank
rpoisson-small	1,961			1,977	3,954	35,450	1,957,230
rpoisson-middle	3,922	1	16	3,938	7,876	70,748	7,759,829
rpoisson-large	19,609			19,625	39,282	353,114	
volatility-small	262			1,605	3,210	59,544	1,290,420
volatility-middle	522	6	33	3,165	6,210	118,044	4,825,170
volatility-large	2,579			15,507	31,014	580,869	
irt-small	3,348			3,541	7,082	671,774	6,274,652
irt-middle	6,695	1	193	6,888	13,776	1,324,439	23,732,604
irt-large	33,475			33,668	67,336	6,546,539	

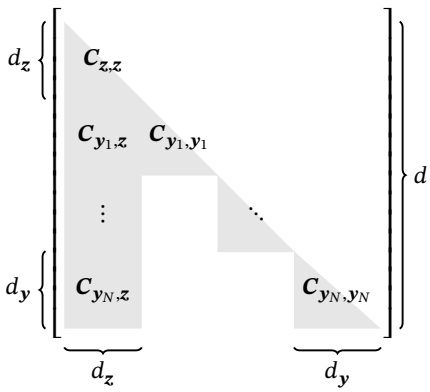

Figure 1: **Visualization of the proposed structure.** The colored entries are non-zero, while the white entries are filled with zeros.

Then, the row-wise blocks of the scale matrix are structured as follows:

$$\boldsymbol{C}_{\boldsymbol{z}} = \begin{bmatrix}\boldsymbol{C}_{\boldsymbol{z},\boldsymbol{z}} & \boldsymbol{0} & \dots & \boldsymbol{0}\end{bmatrix}$$
$$\boldsymbol{C}_{\boldsymbol{y}_n} = \begin{bmatrix}\boldsymbol{C}_{\boldsymbol{y}_n,\boldsymbol{z}} & \boldsymbol{0} & \dots & \boldsymbol{0} & \boldsymbol{C}_{\boldsymbol{y}_n,\boldsymbol{y}_n} & \boldsymbol{0} & \dots & \boldsymbol{0}\end{bmatrix}.$$

The full structure of $\boldsymbol{C}$ is visualized in Figure 1. The key is the structure of the row-wise blocks for the local variables, $\boldsymbol{C}_{\boldsymbol{y}_1}, \dots, \boldsymbol{C}_{\boldsymbol{y}_N}$. The rectangular $\boldsymbol{C}_{\boldsymbol{y}_n,\boldsymbol{z}}$ blocks model the conditional dependence of the local variables on the global variables, while the lower triangular $\boldsymbol{C}_{\boldsymbol{y}_n,\boldsymbol{y}_n}$ model the correlations within each local variables.

Remark 9. The proposed structure has a space/storage complexity of $\mathcal{O}\left(\left(d_{\boldsymbol{z}} d_{\boldsymbol{y}} + d_{\boldsymbol{y}}^2\right) N\right)$. This improves over full-rank, which has a storage complexity of $\mathcal{O}\left(\left(d_{\boldsymbol{y}} + d_{\boldsymbol{z}} N\right)^2\right)$.

Structured Variational Family Perspective By the property of triangular matrices, this implicitly forms a *structured* variational family of the form of

$$q\left(\boldsymbol{z}, \boldsymbol{y}_1, \dots, \boldsymbol{y}_N\right) = q\left(\boldsymbol{z}\right) \prod_{n=1}^{N} q\left(\boldsymbol{y}_n \mid \boldsymbol{z}\right),$$

where each $\boldsymbol{y}_n, \boldsymbol{y}_{n'}$ for $n \neq n'$ are independent conditional on $\boldsymbol{z}$. For example, when φ is chosen to be Gaussian, q becomes

$$q\left(\boldsymbol{z}, \boldsymbol{y}_1, \dots, \boldsymbol{y}_N\right) = \mathcal{N}\left(\boldsymbol{z};\ \boldsymbol{m}_{\boldsymbol{z}},\ \boldsymbol{C}_{\boldsymbol{z},\boldsymbol{z}} \boldsymbol{C}_{\boldsymbol{z},\boldsymbol{z}}^\top\right)$$
$$\times \prod_{n=1}^{N} \mathcal{N}\left(\boldsymbol{y}_n;\ \boldsymbol{m}_{\boldsymbol{y}_n} + \boldsymbol{C}_{\boldsymbol{y}_n,\boldsymbol{z}} \boldsymbol{C}_{\boldsymbol{z},\boldsymbol{z}}^{-1}\left(\boldsymbol{z} - \boldsymbol{m}_{\boldsymbol{z}}\right),\ \boldsymbol{C}_{\boldsymbol{y}_n,\boldsymbol{y}_n} \boldsymbol{C}_{\boldsymbol{y}_n,\boldsymbol{y}_n}^\top\right). \tag{4}$$

Note that we do not actually compute $\boldsymbol{C}_{\boldsymbol{z},\boldsymbol{z}}^{-1}$, and only use the sparse matrix representation in Figure 1 for our experiments. Also, Equation (4) is identical to the structure proposed by Tan (2021, §3) and Tan et al. (2020). However, they did not consider the scalability aspect of this parameterization. Equation (4) is displayed for

A similar structure is also considered by Agrawal and Domke (2021, Table 1):

$$q\left(\boldsymbol{z}, \boldsymbol{y}_1, \dots, \boldsymbol{y}_N\right) = \mathcal{N}\left(\boldsymbol{z};\ \boldsymbol{m}_{\boldsymbol{z}},\ \boldsymbol{C}_{\boldsymbol{z},\boldsymbol{z}} \boldsymbol{C}_{\boldsymbol{z},\boldsymbol{z}}^\top\right)$$
$$\times \prod_{n=1}^{N} \mathcal{N}\left(\boldsymbol{y}_n;\ \boldsymbol{m}_{\boldsymbol{y}_n} + \boldsymbol{A}_n \boldsymbol{z},\ \boldsymbol{C}_{\boldsymbol{y}_n,\boldsymbol{y}_n} \boldsymbol{C}_{\boldsymbol{y}_n,\boldsymbol{y}_n}^\top\right), \tag{5}$$

where $\boldsymbol{A}_n$ is also an optimized parameter. This is equivalent to our representation, in which we whiten $\boldsymbol{z}$ first instead. However, the consequences of the parameterizations are quite different: In Equation (4), solving for $\boldsymbol{C}_{\boldsymbol{y}_n,\boldsymbol{z}}$ does not depend on the parameters of the global variables, $\boldsymbol{m}_{\boldsymbol{z}}, \boldsymbol{C}_{\boldsymbol{z},\boldsymbol{z}}$. Equation (5), on the other hand, introduces a coupling between the parameters of $\boldsymbol{y}_n$ and $\boldsymbol{z}$, making optimization more challenging.

Given Figure 1, we have:

Proposition 1. *The effective dimensionality d^* of the scale matrix structure in Figure 1 is given as*

$$d^* = d_{\boldsymbol{z}} + d_{\boldsymbol{y}}$$

Proof. By its definition in Theorem 1 and our covariance structure, we have

$$\begin{aligned} d^* &= \max_{b=1,\dots,B} \sum_{b'=1}^{B} r_{\boldsymbol{z}_b,\boldsymbol{z}_{b'}} \\ &= \max\left\{\max_{n=1,\dots,N} r_{\boldsymbol{y}_n,\boldsymbol{z}} + r_{\boldsymbol{y}_n,\boldsymbol{y}_n},\ r_{\boldsymbol{z},\boldsymbol{z}}\right\} \\ &= \max\left(d_{\boldsymbol{z}} + d_{\boldsymbol{y}},\ d_{\boldsymbol{z}}\right) = d_{\boldsymbol{z}} + d_{\boldsymbol{y}}. \end{aligned}$$

□

Corollary 2. *Let the assumptions of Theorem 2 hold and the structured variational family in Figure 1 be used. Then, the iteration complexity of finding an ϵ-accurate solution $\boldsymbol{\lambda}_{T+1}$ satisfying $\|\boldsymbol{\lambda}_{T+1} - \boldsymbol{\lambda}^*\|_2^2 \leq \epsilon$ using BBVI*

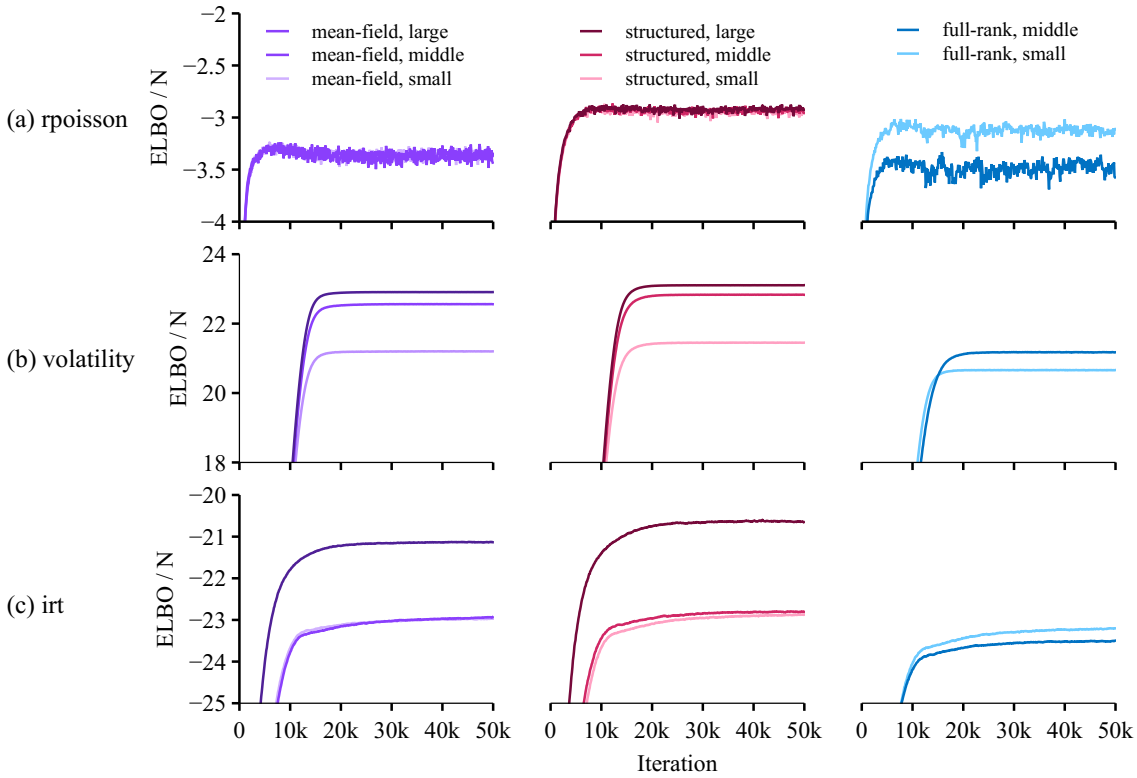

Figure 2: **ELBO normalized by the number of datapoints versus the number of iterations (T) on the considered problems on datasets of varying sizes.** The format of the labels is "¡variational family¿, ¡dataset size¿". *e.g.*, "mean-field, small" on rpoisson is the result for the mean-field family on the rpoisson-small dataset. We are showing the results for a stepsize of $\gamma = 10^{-3.5}$ for rpoisson and irt, and $\gamma = 10^{-3}$ for volatility. The solid lines are the median over 4 independent replications.

with proximal SGD is

$$\mathcal{O}\left(\kappa^2 \left(d_{\boldsymbol{z}} + d_{\boldsymbol{y}} + k_{\varphi}\right) \frac{1}{\epsilon} \log\left(2\Delta_0^2 \frac{1}{\epsilon}\right)\right).$$

Remark 10. Using our proposed covariance structure for the scale matrix, the iteration complexity of BBVI is independent of the number of data points N.

4 EXPERIMENTS

We now evaluate the accuracy of our theoretical analysis in Section 3. Mainly, we will compare the scalability of mean-field, full-rank, and the structured variational family in Figure 1.

4.1 Setup

Stochastic Optimization While our theoretical results use proximal SGD, we use regular SGD without projection for our experiments. In practice, proximal SGD does not provide significant benefit (Kim et al., 2023b), and simple SGD is the most common method for performing BBVI in practice. Also, we only run SGD with fixed stepsizes. In practice, step-decay stepsize schedules Goffin 1977 are believed to be superior to polynomial decay schedules or naive fixed stepsizes. However, the theoretical evidence for this is not yet clear. (See (Ge et al., 2019; Wang et al., 2021) for some recent works on this.) Also, for the sake of experimentation, step-decay stepsizes are hard to control due to the additional number of configurations, such as the number of stages, amount of decay per stage, and such.

Implementation We implemented our experiments using the Turing ecosystem (Ge et al., 2018) in the Julia language (Bezanson et al., 2017). The structured covariances were implemented using the compressed sparse column (CSC) sparse matrix interface provided by the `CUDA.jl` library (Besard et al., 2019), while the sparse derivatives were implemented using the `Zygote.jl` framework (Innes, 2018). We use 8 Monte Carlo samples and the Adam optimizer (Kingma and Ba, 2015) for all problems, while the reported ELBOs are estimated using 1024 Monte Carlo samples every 100 iterations. The variational families are Gaussian such that $\varphi = \mathcal{N}(0, 1)$.

Models and Dataset We use three different Hierarchical Bayesian models, volatility, rpoisson, and irt, which have local variables. rpoisson is a robust generalized linear regression model known as the Poisson-log-normal model (Cameron and Trivedi, 2013, §4.2.4). We treat the regression coefficients and the hyperparameters as global variables, while the individual-level response, which is

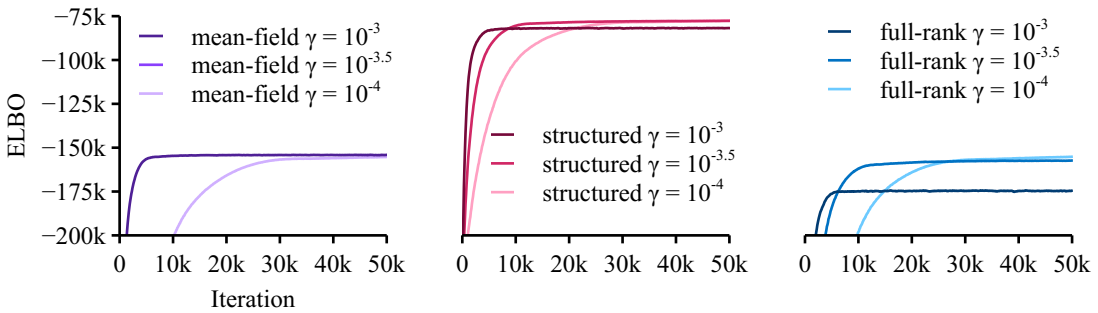

Figure 3: **ELBO depending on the Adam optimizer stepsize (γ) on the irt-small problem.** The solid lines are the median over 4 independent replications.

modeled to include log-normal noise, is treated as the local variable. volatility is a multivariate stochastic volatility model (Chib et al., 2009; Naesseth et al., 2018). We treat the hyperparameters as the global variables, while the latent volatilities are treated as the local variables. Lastly, irt is a two-parameter logistic item response theory (IRT) model (Lord et al., 2008; Wu et al., 2020). We treat the "ability" of each student as local variables, while the hyperparameters and item-level variables are treated as global variables. The full description of these models can be found in Appendix C. To evaluate the effect of dataset size, we use subsets of the full datasets, as shown in Table 1. For the initial point, we use $q_{\lambda_0} = \mathcal{N}\left(\mathbf{0}, 10^{-2}\mathbf{I}\right)$ for all experiments.

4.2 Results

We now present the experimental results. Here, we present refined results for interpretability. The full set of results, including confidence intervals, is in Appendix D.

Scaling with Respect to N The effect of scaling up the dataset is illustrated in Figure 2. Since the size of the dataset affects the scale of the ELBOs, we normalized them by the number of data points to put them on a comparable scale. On all problems, the structured family results in the highest normalized ELBOs regardless of the N. This confirms that structured families more accurately approximate the posterior than mean-field families. Furthermore, on volatility and irt, structured takes advantage of the larger N, resulting in higher nats per datapoint. This contrasts with full-rank and mean-field, which do not always improve with N. Therefore, the experimental results support the theoretical results that structured families can scale.

Gradient Variance and SGD Stepsize Notice that full-rank *worsens* with more data on rpoisson and irt. The theory suggests that this is due to gradient noise, which, in principle, can be combated by using a smaller stepsize at the price of a larger number of iterations T.

The effect of changing the stepsize is presented in Figure 3. For the structured and full-rank variational families, changing the stepsize has a clear effect on the solution found by Adam. This exactly agrees with the theory: for fixed stepsize SGD, we trade-off between optimization speed and the accuracy of the solution. Larger gradient variance makes this trade-off more noticeable in the full-rank family. While the structured family is also affected by the stepsize, we can see that the effect is smaller than the full-rank family.

5 DISCUSSIONS

Conclusions In this work, we have theoretically investigated the limitations of full-rank variational families in BBVI. Specifically, the dimensional scaling of the gradient variance of full-rank variational families. This is particularly problematic for Bayesian models with local variables implying that BBVI with full-rank variational families do not scale to larger datasets. Fortunately, we have rigorously shown that variational families with structured covariances are able to improve this scaling issue. We have evaluated this theoretical insight on large-scale Hierarchical Bayesian models and have confirmed that practice agrees with the theory. Furthermore, our theoretical analysis provides a precise quantitative analysis of how certain covariance structures would improve the computational complexity of BBVI.

Related Works Structured variational approximations have a long history since their use in hidden Markov models (Ghahramani and Jordan, 1997; Saul and Jordan, 1995), CAVI (Hoffman and Blei, 2015; Mimno et al., 2012; Rohde and W, 2016), fixed-form Variational Bayes (Salimans and Knowles, 2013), BBVI (Ong et al., 2018; Quiroz et al., 2023; Ranganath et al., 2016; Tan, 2021; Tan et al., 2020), and now modern approaches such as amortized VI (Agrawal and Domke, 2021; Archer et al., 2015; Gao et al., 2016; Johnson et al., 2016; Maaløe et al., 2016; Sø nderby et al., 2016; Webb et al., 2018; Yu et al., 2022) and normalizing flows (Ambrogioni et al., 2021a,b). Our proposed covariance structure is identical to that of Tan (2021); Tan et al. (2020). But, they did not consider the computational complexity of this parameterization, for which we rigorously prove its scalability with respect to N.

The computational theory of BBVI is accurate. For full-rank families, we have also evaluated the remedy provided by stochastic optimization theory: reduce the stepsize and run SGD longer. Indeed, the experimental results confirm that this improves the resulting ELBO. Therefore, this

demonstrates that the theoretical framework on the computational theory of BBVI established by Domke (2019, 2020); Domke et al. (2023); Kim et al. (2023b,c) is useful to analyze and improve BBVI.

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Checklist

1. For all models and algorithms presented, check if you include:
   (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes/No/Not Applicable]
   Yes.
   (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes/No/Not Applicable]
   Yes. We discuss the time complexity.
   (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes/No/Not Applicable]
   No.

2. For any theoretical claim, check if you include:
   (a) Statements of the full set of assumptions of all theoretical results. [Yes/No/Not Applicable]
   Yes.
   (b) Complete proofs of all theoretical results. [Yes/No/Not Applicable]
   Yes, in the supplementary material.
   (c) Clear explanations of any assumptions. [Yes/No/Not Applicable]
   Yes, in the main text and the supplementary material.

3. For all figures and tables that present empirical results, check if you include:
   (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes/No/Not Applicable]
   No. But, we provide written details on the implementation and the datasets. All of the datasets are publicly available.
   (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes/No/Not Applicable]
   Yes.
   (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes/No/Not Applicable]
   Yes. The error bars have been omitted in the main text, but are presented in the extended results in Appendix D.
   (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes/No/Not Applicable]
   Yes.

4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
   (a) Citations of the creator If your work uses existing assets. [Yes/No/Not Applicable]
   Yes.
   (b) The license information of the assets, if applicable. [Yes/No/Not Applicable]
   No.
   (c) New assets either in the supplemental material or as a URL, if applicable. [Yes/No/Not Applicable]
   Not Applicable.
   (d) Information about consent from data providers/curators. [Yes/No/Not Applicable]
   Not Applicable.
   (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Yes/No/Not Applicable]
   Not Applicable.

5. If you used crowdsourcing or conducted research with human subjects, check if you include:
   (a) The full text of instructions given to participants and screenshots. [Yes/No/Not Applicable]
   Not Applicable.
   (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Yes/No/Not Applicable]
   Not Applicable.
   (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Yes/No/Not Applicable]
   Not Applicable.

Provably Scalable Black-Box Variational Inference with Structured Covariances

Supplementary Material

A COMPUTATIONAL RESOURCES

Table 2: Computational Resources

Type	Model and Specifications
System Topology	1 socket with 8 physical cores
Processor	1 Intel i9-11900F, 2.5 GHz (maximum 5.2 GHz) per socket
Cache	80 KB L1, 512 KB L2, and 16 MB L3
Memory	64 GiB RAM
Accelerator	1 NVIDIA GeForce RTX 3090 per node, 1.7 GHZ, 24GiB RAM

All of the experiments took approximately 1 week to run.

B PROOFS

B.1 Working with Blocks of Coordinates

In this work, we need to analyze the properties of block partitions of the coordinates. For this, we use the following notation:

Definition 2 (Block Identity Matrix). Consider a block partitioning of $\boldsymbol{z} \in \mathbb{R}^d$ as $\boldsymbol{z} = (\boldsymbol{z}_1, \dots, \boldsymbol{z}_b, \dots, \boldsymbol{z}_B)$, where $\boldsymbol{z}_b \in \mathbb{R}^{d_{\boldsymbol{z}_b}}$ such that $d = \sum_{b=1}^{B} d_{\boldsymbol{z}_b}$. Then, $\mathbf{I}_{\boldsymbol{z}_b} \in \mathbb{R}^{d \times d}$ is a diagonal matrix where the diagonals corresponding to the coordinates of $\boldsymbol{z}_b$ are set to 1.

It is straightforward that $\mathbf{I}_{\boldsymbol{z}_b}$ acts as a *unit basis* for the space of $d \times d$ diagonal matrices such that $\sum_{b=1}^{B} \mathbf{I}_{\boldsymbol{z}_b} = \mathbf{I}$. Furthermore, consider some arbitrary vector

$$\boldsymbol{x} = (\boldsymbol{x}_1, \dots, \boldsymbol{x}_b, \dots, \boldsymbol{x}_B),$$

where $\boldsymbol{x}_b \in \mathbb{R}^{d_{\boldsymbol{z}_b}}$ such that the length of $\boldsymbol{x}_b$ and $\boldsymbol{z}_b$ is equal. Then, we have the following properties:

$$\boldsymbol{x} = \sum_{b=1}^{B} \mathbf{I}_{\boldsymbol{z}_b} \boldsymbol{x},$$

$$\|\boldsymbol{x}_b\| = \|\mathbf{I}_{\boldsymbol{z}_b} \boldsymbol{x}\|_2^2 = \boldsymbol{x}_b^\top \mathbf{I}_{\boldsymbol{z}_b} \boldsymbol{x}_b.$$

These properties are straightforward to check, so we present them without proof.

Also, for notational clarity, we define the following:

$$\nabla_{\boldsymbol{z}_b} f(\boldsymbol{z}) \triangleq \mathbf{I}_{\boldsymbol{z}_b} \nabla f(\boldsymbol{z})$$

for any differentiable function $f : \mathbb{R}^d \to \mathbb{R}$.

B.2 Definitions

Definition (L**-Smoothness**)**.** A function $f : \mathcal{X} \to \mathbb{R}$ is L-smooth if it satisfies

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\|_2 \leq L\|\boldsymbol{x} - \boldsymbol{y}\|_2$$

for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{X}$ and some $L > 0$.

Remark 11. Equivalently, we say a function f is L-log-smooth if $\log f$ is L-smooth.

Definition (μ**-Strong Convexity**)**.** A function $f : \mathcal{X} \to \mathbb{R}$ is μ-strongly convex if it satisfies

$$\frac{\mu}{2}\|\boldsymbol{x} - \boldsymbol{y}\|_2^2 \leq f(\boldsymbol{y}) - f(\boldsymbol{x}) - \langle \nabla f(\boldsymbol{x}), \boldsymbol{y} - \boldsymbol{x} \rangle$$

for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{X}$ and some $\mu > 0$.

Remark 12. Equivalently, we say a function f is only convex if it satisfies the strong convexity inequality with $\mu = 0$.

Remark 13 (**Log-Concave Measures**)**.** For a probability measure Π, we say it is μ-strongly log-concave if, in a d-dimensional Euclidean measurable space $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \mathbb{P})$, where $\mathcal{B}(\mathbb{R}^d)$ is the σ-algebra of Borel-measurable subsets of $\mathbb{R}^d$ and $\mathbb{P}$ is the Lebesgue measure, its log probability density function $x \mapsto -\log \pi(x)$ is μ-strongly convex.

B.3 Auxiliary Lemmas

Lemma 2. *Let $\boldsymbol{u} = (u_1, u_2, \dots, u_d)$ be a d-dimensional vector-valued random variable satisfying Assumption 1. Then,*

1. $\mathbb{E}\boldsymbol{u}\boldsymbol{u}^\top = \mathbf{I}$
2. $\mathbb{E}u_i^2\,\boldsymbol{u} = \mathbf{0}$,

for any arbitrary $i = 1, \dots, d$.

Proof. The first identity is derived by Domke (2019, Lemma 9).

$$\mathbb{E}u_i^2\,\boldsymbol{u} = \mathbb{E}\begin{bmatrix} u_1 & \dots & u_{i-1} & u_i^3 & u_{i+1} & \dots & u_d \end{bmatrix}^\top = \begin{bmatrix} \mathbb{E}u_1 & \dots & \mathbb{E}u_{i-1} & \mathbb{E}u_i^3 & \mathbb{E}u_{i+1} & \dots & \mathbb{E}u_d \end{bmatrix}^\top = \mathbf{0},$$

where the last equality follows from Assumption 1. □

Lemma 3. *Let $\boldsymbol{A}$ be some diagonal matrix formed as $\boldsymbol{A} = \sum_{b=1}^{B} c_b \mathbf{I}_{\boldsymbol{z}_b}$ and $\boldsymbol{z} \in \mathbb{R}^d$ be block-structured as $\boldsymbol{z} = (\boldsymbol{z}_1, \dots, \boldsymbol{z}_b, \dots, \boldsymbol{z}_B)$. Then, a quadratic form with respect to $\boldsymbol{A}$ satisfies*

$$\boldsymbol{z}^\top \boldsymbol{A}\boldsymbol{z} = \sum_{b=1}^{B} c_b \|\boldsymbol{z}_b\|_2^2,$$

for any vector $\boldsymbol{z} \in \mathbb{R}^d$.

Proof.

$$\boldsymbol{z}^\top \boldsymbol{A}\boldsymbol{z} = \boldsymbol{z}^\top \left(\sum_{b=1}^{B} c_b \mathbf{I}_{\boldsymbol{z}_b} \right) \boldsymbol{z} = \sum_{b=1}^{B} c_b \left(\boldsymbol{z}^\top \mathbf{I}_{\boldsymbol{z}_b} \boldsymbol{z} \right) = \sum_{b=1}^{B} c_b \|\boldsymbol{z}_b\|_2^2.$$

□

Lemma 4. *Let $\boldsymbol{z} = (\boldsymbol{z}_1, \dots, \boldsymbol{z}_b, \dots, \boldsymbol{z}_B) \in \mathbb{R}^d$ be a block partition of the coordinates of an L-smooth differentiable function $f : \mathbb{R}^d \to \mathbb{R}$. Then, f is also L-block-wise smooth such that*

$$\left\|\nabla_{\boldsymbol{z}_b} f\left(\boldsymbol{x} + \mathbf{I}_{\boldsymbol{z}_b}\boldsymbol{d}\right) - \nabla_{\boldsymbol{z}_b} f\left(\boldsymbol{x}\right)\right\| \leq L\|\boldsymbol{d}\|$$

for any non-overlapping block partitioning and any block b and any $\boldsymbol{x}, \boldsymbol{d} \in \mathbb{R}^d$.

Proof. From L-smoothness we have

$$\|\nabla f\left(\boldsymbol{x}\right) - \nabla f\left(\boldsymbol{y}\right)\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|.$$

Considering this,

$$\begin{aligned}\left\|\nabla f\left(\boldsymbol{x} + \mathbf{I}_{\boldsymbol{z}_b}\boldsymbol{d}\right) - \nabla f\left(\boldsymbol{x}\right)\right\| &= \left\|\mathbf{I}_{\boldsymbol{z}_b}\left(\nabla f\left(\boldsymbol{x} + \mathbf{I}_{\boldsymbol{z}_b}\boldsymbol{d}\right) - \nabla f\left(\boldsymbol{x}\right)\right) + \underbrace{\left(\mathbf{I} - \mathbf{I}_{\boldsymbol{z}_b}\right)\left(\nabla f\left(\boldsymbol{x} + \mathbf{I}_{\boldsymbol{z}_b}\boldsymbol{d}\right) - \nabla f\left(\boldsymbol{x}\right)\right)}_{=0}\right\| \\ &= \left\|\mathbf{I}_{\boldsymbol{z}_b}\left(\nabla f\left(\boldsymbol{x} + \mathbf{I}_{\boldsymbol{z}_b}\boldsymbol{d}\right) - \nabla f\left(\boldsymbol{x}\right)\right)\right\|.\end{aligned}$$

The last equality follows because the coordinates of $\nabla f\left(\boldsymbol{x} + \mathbf{I}_{\boldsymbol{z}_b}\boldsymbol{d}\right) - \nabla f\left(\boldsymbol{x}\right)$ not masked by $\mathbf{I}_{\boldsymbol{z}_b}$ are not being perturbed since we also mask the perturbation $\boldsymbol{d}$.

Since we also have

$$\left\|\nabla f\left(\boldsymbol{x} + \mathbf{I}_{\boldsymbol{z}_b}\boldsymbol{d}\right) - \nabla f\left(\boldsymbol{x}\right)\right\| \leq L\|\boldsymbol{d}\|,$$

it also follows that

$$\left\|\mathbf{I}_{\boldsymbol{z}_b}\left(\nabla f\left(\boldsymbol{x} + \mathbf{I}_{\boldsymbol{z}_b}\boldsymbol{d}\right) - \nabla f\left(\boldsymbol{x}\right)\right)\right\| = \left\|\nabla_{\boldsymbol{z}_b} f\left(\boldsymbol{x} + \mathbf{I}_{\boldsymbol{z}_b}\boldsymbol{d}\right) - \nabla_{\boldsymbol{z}_b} f\left(\boldsymbol{x}\right)\right\| \leq L\|\boldsymbol{d}\|.$$

□

Lemma 5 (Rammus, 2021)**.** *Consider $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{d\times d}$, where $\boldsymbol{B}$ is positive definite such that $\boldsymbol{B} \succ 0$. Then,*

$$\operatorname{tr}\left(\boldsymbol{A}\boldsymbol{B}\right) \leq \|\boldsymbol{A}\|_{2,2}\operatorname{tr}\left(\boldsymbol{B}\right).$$

Proof. We restate the proof of Rammus (2021) for completeness.

Consider an inner product space of matrices, where, for $\boldsymbol{X}, \boldsymbol{Y} \in \mathbb{C}^{d\times d}$, its inner product is defined as $\langle \boldsymbol{X}, \boldsymbol{Y}\rangle = \operatorname{tr}\left(\boldsymbol{X}^*\boldsymbol{Y}\right)$. Here, $\boldsymbol{X}^*$ is the conjugate transpose of $\boldsymbol{X}$. This generates a p-norm as

$$\|\boldsymbol{X}\|_p \triangleq \left(\sum_i \sigma_i\left(\boldsymbol{X}\right)^p\right)^{1/p},$$

where $\sigma_i\left(\boldsymbol{X}\right)$ is the ith singular value of $\boldsymbol{X}$. This norm is known as the Schatten norm, where its limiting case $p \to \infty$ is, in fact, the ℓ_2 operator norm.

We can now apply Hölder's inequality

$$|\langle \boldsymbol{A}, \boldsymbol{B}\rangle| \leq \|\boldsymbol{A}\|_p\|\boldsymbol{B}\|_q,$$

which is valid for $\frac{1}{p} + \frac{1}{q} = 1$. By choosing $p \to \infty$ and $q \to 1$, we have

$$\operatorname{tr}\left(\boldsymbol{A}\boldsymbol{B}\right) = \left\langle \boldsymbol{A}^\top, \boldsymbol{B}\right\rangle \leq \|\boldsymbol{A}\|_\infty\|\boldsymbol{B}\|_1 = \|\boldsymbol{A}\|_{2,2}\|\boldsymbol{B}\|_1 = \|\boldsymbol{A}\|_\infty\operatorname{tr}\left(\boldsymbol{B}\right),$$

where the last equality follows from the fact that, for positive definite matrices such as $\boldsymbol{B}$, it follows that

$$\operatorname{tr}\left(\boldsymbol{B}\right) = \sum_i \lambda_i\left(\boldsymbol{B}\right) = \sum_i \sigma_i\left(\boldsymbol{B}\right),$$

where $\lambda_i\left(\boldsymbol{B}\right)$ is the ith eigenvalue of $\boldsymbol{B}$.

□

B.4 Gradient Variance Upper Bound

B.4.1 Key Lemmas

Lemma 6. *Let $\boldsymbol{u}$ be block structured as*

$$\boldsymbol{u} = \left(\boldsymbol{u}_{z_1}, \ \dots, \ \boldsymbol{u}_{z_b}, \ \dots, \ \boldsymbol{u}_{z_B}\right),$$

where each block is $\boldsymbol{u}_{z_b} \in \mathbb{R}^{d_{z_b}}$ such that $\sum_{b=1}^{B} d_{z_b} = d$. The squared Jacobian of the reparameterization function $\mathcal{T} : \mathbb{R}^d \times \Lambda \to \mathbb{R}^d$ is a diagonal matrix given as

$$\left(\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{\lambda}}\right)^\top \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{\lambda}} \preceq \mathbf{I} + \sum_{b=1}^{B} \left(\sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j} \, u_{z_{b'}, j}^2 \right) \mathbf{I}_{z_b},$$

where

$$\delta_{z_b, z_{b'}, j} = \begin{cases} 1 & \text{if the } j\text{th column of the block } \boldsymbol{C}_{z_b, z_{b'}} \text{ is non-zero} \\ 0 & \text{otherwise} \end{cases}$$

and $u_{z_{b'}, j}$ is the jth element of the block of $\boldsymbol{u}$ corresponding to $\boldsymbol{z}_{b'}$.

Proof. From the definition of the location-scale reparameterization function, we have

$$\mathcal{T}_\lambda(\boldsymbol{u}) = \boldsymbol{C}\boldsymbol{u} + \boldsymbol{m}.$$

The Jacobian is a block-structured matrix that can be described as

$$\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{\lambda}} = \begin{bmatrix} \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{m}} & \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{C}_{z_1}} & \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{C}_{z_2}} & \dots & \frac{\partial \mathcal{T}_n(\boldsymbol{u};\lambda)}{\partial \boldsymbol{C}_{z_{B-1}}} & \frac{\partial \mathcal{T}_n(\boldsymbol{u};\lambda)}{\partial \boldsymbol{C}_{z_B}} \end{bmatrix}^\top,$$

where $\boldsymbol{C}_{z_b}$ is the bth row-wise block of $\boldsymbol{C}$. Now,

$$\begin{aligned} \left(\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{\lambda}}\right)^\top \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{\lambda}} &= \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{m}} \left(\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{m}}\right)^\top + \sum_{b=1}^{B} \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{C}_{z_b}} \left(\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{C}_{z_b}}\right)^\top \\ &= \mathbf{I} + \sum_{b=1}^{B} \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{C}_{z_b}} \left(\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{C}_{z_b}}\right)^\top. \end{aligned}$$

Recall that we have defined $\boldsymbol{C}_{z_b}$ as

$$\boldsymbol{C}_{z_b} = \begin{bmatrix} \boldsymbol{C}_{z_b, z_1} & \dots & \boldsymbol{C}_{z_b, z_b} & \mathbf{0} & \dots & \mathbf{0} \end{bmatrix}.$$

Then,

$$\begin{aligned} \left(\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{\lambda}}\right)^\top \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{\lambda}} &= \mathbf{I} + \sum_{b=1}^{B} \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{C}_{z_b}} \left(\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{C}_{z_b}}\right)^\top \\ &= \mathbf{I} + \sum_{b=1}^{B} \sum_{b'=1}^{B} \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{C}_{z_b, z_{b'}}} \left(\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{C}_{z_b, z_{b'}}}\right)^\top. \end{aligned}$$

For the derivatives, Domke (2019, Lemma 8) show that

$$\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial m_i} = \mathbf{e}_i \quad \text{and} \quad \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial C_{i,j}} = \mathbf{e}_i u_j,$$

where $\mathbf{e}_i$ is the unit bases for the ith coordinate. Here, we are interested in the effect of the structure of $\boldsymbol{C}$.

Using the indicator δ, we have

$$\frac{\partial \mathcal{T}(\boldsymbol{u};\boldsymbol{\lambda})}{\partial C_{z_b, z_{b'}, i, j}} = \delta_{z_b, z_{b'}, i, j} \, \mathbf{e}_{z_b, i} \, u_{z_{b'}, j}, \tag{6}$$

where is an indicator of whether an element of $\boldsymbol{C}$ is zero or not, $u_{z_{b'},j}$ is the jth element of the block of $\boldsymbol{u}$ corresponding to $z_{b'}$, $\mathbf{e}_{z_b,i}$ is the unit bases for the ith element of the block corresponding to z_b and

$$\delta_{z_b,z_{b'},i,j} = \begin{cases} 1 & \text{if the } i,j\text{th element of the block } \boldsymbol{C}_{z_b,z_{b'}} \text{ is non-zero} \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,

$$\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial C_{z_b,z_{b'},i,j}} \left(\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial C_{z_b,z_{b'},i,j}} \right)^\top = \left(\delta_{z_b,z_{b'},i,j}\, \mathbf{e}_{z_b,i}\, u_{z_{b'},j} \right) \left(\delta_{z_b,z_{b'},i,j}\, \mathbf{e}_{z_b,i}\, u_{z_{b'},j} \right)^\top = \delta_{z_b,z_{b'},i,j}\, u^2_{z_{b'},j} \left(\mathbf{e}_{z_b,i}\, \mathbf{e}^\top_{z_b,i} \right),$$

and thus

$$\begin{aligned} \left(\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \lambda} \right)^\top \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \lambda} &= \mathbf{I} + \sum_{b=1}^{B} \sum_{b'=1}^{B} \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{C}_{z_b,z_{b'}}} \left(\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \boldsymbol{C}_{z_b,z_{b'}}} \right)^\top \\ &= \mathbf{I} + \sum_{b=1}^{B} \sum_{b'=1}^{B} \sum_i \sum_j \delta_{z_b,z_{b'},i,j}\, u^2_{z_{b'},j} \left(\mathbf{e}_{z_b,i}\, \mathbf{e}^\top_{z_b,i} \right). \end{aligned}$$

Here, notice the sum over b, i forms a diagonal matrix since $\mathbf{e}_{z_b,i}\, \mathbf{e}^\top_{z_b,i}$ is the unit basis for diagonal matrices. In fact, this is a positive semi-definite matrix since the diagonal entries are sums of non-negative elements. Therefore, we can upper-bound the matrix in the Loewner order. For this, we will define a new indicator

$$\delta_{z_b,z_{b'},j} = \begin{cases} 1 & \text{if } j\text{the column of } \boldsymbol{C}_{z_b,z_{b'}} \text{ is non-zero} \\ 0 & \text{otherwise.} \end{cases}$$

That is, $\delta_{z_b,z_{b'},i,j} = 1$ if $\delta_{z_b,z_{b'},j} = 1$. Applying this,

$$\begin{aligned} \left(\frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \lambda} \right)^\top \frac{\partial \mathcal{T}_\lambda(\boldsymbol{u})}{\partial \lambda} &= \mathbf{I} + \sum_{b'=1}^{B} \sum_j \sum_{b=1}^{B} \sum_i \delta_{z_b,z_{b'},i,j}\, u^2_{z_{b'},j} \left(\mathbf{e}_{z_b,i} \mathbf{e}^\top_{z_b,i} \right) \\ &\leq \mathbf{I} + \sum_{b'=1}^{B} \sum_j \sum_{b=1}^{B} \delta_{z_b,z_{b'},j}\, u^2_{z_{b'},j} \left(\mathbf{e}_{z_b}\, \mathbf{e}^\top_{z_b} \right) \\ &= \mathbf{I} + \sum_{b=1}^{B} \left(\sum_{b'=1}^{B} \sum_j \delta_{z_b,z_{b'},j}\, u^2_{z_{b'},j} \right) \mathbf{I}_{z_b}. \end{aligned}$$

□

Lemma 7. *Let δ_{ij} be an indicator such that $\delta_{ij} \in \{0,1\}$ depending on the index $i, j = 1, \dots, d$. Furthermore, $\boldsymbol{u}$ satisfy Assumption 1 and all expectations be respect to $\boldsymbol{u}$. Then,*

$$\left\| \mathbb{E} \sum_j \delta_{ij}\ u_j^2 \boldsymbol{u}\boldsymbol{u}^\top \right\|_{2,2} \leq \max_i \sum_j \delta_{ij} + k_\varphi - 1,$$

where the equality holds if the values of δ are equal along the columns (j-axis) such that $\delta_{ij} = \delta_{i'j}$ for all i, i'.

Proof. Let us denote

$$\boldsymbol{A} = \mathbb{E} \sum_j \delta_{ij}\ u_j^2 \boldsymbol{u}\boldsymbol{u}^\top$$

for clarity. Then, we have

$$\begin{aligned} A_{ij} &= \mathbb{E} \sum_k \delta_{ik}\ u_k^2\ (\boldsymbol{u}\boldsymbol{u}^\top)_{ij} \\ &= \sum_k \delta_{ik}\ \mathbb{E} u_k^2 u_i u_j. \end{aligned}$$

Especially for the diagonal elements, we have

$$\begin{aligned} A_{ii} &= \sum_k \delta_{ik}\ \mathbb{E} u_k^2 u_i^2 \\ &= \sum_{k \neq i} \delta_{ik}\ \mathbb{E} u_k^2 u_i^2 + \delta_{ii} \mathbb{E} u_i^4 \\ &= \sum_{k \neq i} \delta_{ik}\ + \delta_{ii}\, k_\varphi && \text{(Assumption 1)} \\ &\leq \sum_{k \neq i} \delta_{ik}\ + \delta_{ii}\, k_\varphi + \underbrace{\left(k_\varphi - \delta_{ii}\, k_\varphi + \delta_{ii} - 1\right)}_{\geq 0 \text{ since } k_\varphi \geq 1} \\ &= \sum_k \delta_{ik} + k_\varphi - 1. \end{aligned}$$

For the off-diagonal elements,

$$\begin{aligned} A_{ij} &= \sum_k \delta_{ik}\ \mathbb{E} u_k^2 u_i u_j \\ &= \mathbb{E} \left(\sum_{(k \neq i) \wedge (k \neq j)} \delta_{ik}\ u_k^2 u_i u_j \right) + \mathbb{E} \left(\delta_{ii}\ u_i^2 u_i u_j \right) + \mathbb{E} \left(\delta_{ij}\ u_j^2 u_j u_j \right) \\ &= \left(\sum_{(k \neq i) \wedge (k \neq j)} \delta_{ik}\ \mathbb{E} u_k^2 \mathbb{E} u_i \mathbb{E} u_j \right) + \mathbb{E} \left(\delta_{ii}\ \mathbb{E} u_i^3 \mathbb{E} u_j \right) + \mathbb{E} \left(\delta_{ij}\ \mathbb{E} u_j^3 \mathbb{E} u_j \right) \\ &= 0. && \text{(Assumption 1)} \end{aligned}$$

Therefore, $\boldsymbol{A}$ is a diagonal matrix and the ℓ_2 matrix norm follows as the maximal diagonal element such that

$$\|\boldsymbol{A}\|_{2,2} \leq \max_i \sum_j \delta_{ij} + k_\varphi - 1.$$

□

Lemma 8. *Let $\ell : \mathbb{R}^d \to \mathbb{R}$ be a differentiable function and $\mathcal{T}$ be the location-scale reparameterization function. Then, we have*

$$\|\nabla_{\boldsymbol{\lambda}} \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right)\|_2^2 \leq \sum_{b=1}^{B}\left(1 + \sum_{b'=1}^{B}\sum_{j} \delta_{z_b, z_{b'}, j}\; u_{z_{b'}, j}^2\right)\left\|\nabla_{z_b} \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right)\right\|_2^2$$

for any $\boldsymbol{\lambda} \in \Lambda$ and any $\boldsymbol{u} \in \mathbb{R}^d$, where $\delta_{z_b, z_{b'}, j}$ and $u_{z_{b'}, j}$ are defined in Lemma 6.

Proof.

$$\begin{aligned}
\|\nabla_{\boldsymbol{\lambda}} \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right)\|_2^2 &= \left\|\frac{\partial \mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)}{\partial \boldsymbol{\lambda}} \nabla \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right)\right\|_2^2, \\
&= \nabla \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right)^\top \left(\frac{\partial \mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)}{\partial \boldsymbol{\lambda}}\right)^\top \frac{\partial \mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)}{\partial \boldsymbol{\lambda}} \nabla \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right)
\end{aligned}$$

and since this is a quadratic form, by the definition of positive definite matrices, we can apply Lemma 6 as

$$\leq \nabla \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right)^\top \left(\mathbf{I} + \sum_{b=1}^{B}\left(\sum_{b'=1}^{B}\sum_{j} \delta_{z_b, z_{b'}, j}\; u_{z_{b'}, j}^2\right)\mathbf{I}_{z_b}\right) \nabla \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right).$$

Now, using the properties discussed in Appendix B.1, we can block-decompose the quadratic form as

$$\begin{aligned}
&\nabla \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right)^\top \left(\sum_{b=1}^{B} \mathbf{I}_{z_b} + \sum_{b=1}^{B}\left(\sum_{b'=1}^{B}\sum_{j} \delta_{z_b, z_{b'}, j}\; u_{z_{b'}, j}^2\right)\mathbf{I}_{z_b}\right) \nabla \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right) \\
&= \nabla \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right)^\top \left(\sum_{b=1}^{B}\left(1 + \sum_{b'=1}^{B}\sum_{j} \left(\delta_{z_b, z_{b'}, j}\; u_{z_{b'}, j}^2\right)\right)\mathbf{I}_{z_b}\right) \nabla \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right) \\
&= \sum_{b=1}^{B}\left(1 + \sum_{b'=1}^{B}\sum_{j} \delta_{z_b, z_{b'}, j}\; u_{z_{b'}, j}^2\right)\left\|\mathbf{I}_{z_b} \nabla \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right)\right\|_2^2 \\
&= \sum_{b=1}^{B}\left(1 + \sum_{b'=1}^{B}\sum_{j} \delta_{z_b, z_{b'}, j}\; u_{z_{b'}, j}^2\right)\left\|\nabla_{z_b} \ell\left(\mathcal{T}_{\boldsymbol{\lambda}}\left(\boldsymbol{u}\right)\right)\right\|_2^2.
\end{aligned}$$

□

Lemma 9. *Let $\mathcal{T}_\lambda$ be the location-scale reparameterization function where $\boldsymbol{m}$ and $\boldsymbol{C}$ can be block decomposed as in Section 3.2. Also, let the random variable $\boldsymbol{u}$ satisfy Assumption 1 where all expectations are with respect to $\boldsymbol{u}$. Then, we have*

$$\mathbb{E}\left(1+\sum_{b'=1}^{B}\sum_{j}\delta_{z_b,z_{b'},j}\ u^2_{z_{b'},j}\right)\left\|\mathbf{I}_{z_b}\left(\mathcal{T}_\lambda\left(\boldsymbol{u}\right)-\bar{\boldsymbol{z}}\right)\right\|_2^2 \leq \left(\sum_{b'=1}^{B} r_{z_b,z_{b'}}+1\right)\left\|\mathbf{I}_{z_b}\left(\boldsymbol{m}-\bar{\boldsymbol{z}}\right)\right\|_2^2+\left(\sum_{b'=1}^{B} r_{z_b,z_{b'}}+k_\varphi\right)\left\|\mathbf{I}_b\boldsymbol{C}\right\|_{\mathrm{F}}^2$$

for any $\bar{\boldsymbol{z}}\in\mathbb{R}^d$, any $b=1,\ldots,B$, and $\boldsymbol{\lambda}\in\Lambda$, where $r_{z_b,z_{b'}}$ is defined in Equation (3), $\delta_{z_b,z_{b'},j}$ and $u^2_{z_{b'},j}$ are defined in Lemma 6.

Proof. From the definitions, we have

$$\begin{aligned}
&\mathbb{E}\left(1+\sum_{b'=1}^{B}\sum_{j}\delta_{z_b,z_{b'},j}\ u^2_{z_{b'},j}\right)\left\|\mathbf{I}_{z_b}\left(\mathcal{T}_\lambda\left(\boldsymbol{u}\right)-\bar{\boldsymbol{z}}\right)\right\|_2^2\\
&=\mathbb{E}\left(1+\sum_{b'=1}^{B}\sum_{j}\delta_{z_b,z_{b'},j}\ u^2_{z_{b'},j}\right)\left\|\mathbf{I}_{z_b}\left(\boldsymbol{C}\boldsymbol{u}+\boldsymbol{m}-\bar{\boldsymbol{z}}\right)\right\|_2^2\\
&=\mathbb{E}\left(1+\sum_{b'=1}^{B}\sum_{j}\delta_{z_b,z_{b'},j}\ u^2_{z_{b'},j}\right)\left\|\boldsymbol{C}_{z_b}\boldsymbol{u}+\boldsymbol{m}_{z_b}-\bar{\boldsymbol{z}}_{z_b}\right\|_2^2,
\end{aligned}$$

expanding the square,

$$\begin{aligned}
&=\mathbb{E}\left(1+\sum_{b'=1}^{B}\sum_{j}\delta_{z_b,z_{b'},j}\ u^2_{z_{b'},j}\right)\left(\boldsymbol{u}^\top\boldsymbol{C}_{z_b}^\top\boldsymbol{C}_{z_b}\boldsymbol{u}-2\left(\boldsymbol{m}_{z_b}-\bar{\boldsymbol{z}}_{z_b}\right)^\top\boldsymbol{C}_{z_b}\boldsymbol{u}+\left\|\boldsymbol{m}_{z_b}-\bar{\boldsymbol{z}}_{z_b}\right\|_2^2\right)\\
&=\underbrace{\mathbb{E}\left(1+\sum_{b'=1}^{B}\sum_{j}\delta_{z_b,z_{b'},j}\ u^2_{z_{b'},j}\right)\boldsymbol{u}^\top\boldsymbol{C}_{z_b}^\top\boldsymbol{C}_{z_b}\boldsymbol{u}}_{T_b^{(1)}}\\
&\quad-2\underbrace{\mathbb{E}\left(1+\sum_{b'=1}^{B}\sum_{j}\delta_{z_b,z_{b'},j}\ u^2_{z_{b'},j}\right)\left(\boldsymbol{m}_{z_b}-\bar{\boldsymbol{z}}_{z_b}\right)^\top\boldsymbol{C}_{z_b}\boldsymbol{u}}_{T_b^{(2)}}\\
&\quad+\underbrace{\mathbb{E}\left(1+\sum_{b'=1}^{B}\sum_{j}\delta_{z_b,z_{b'},j}\ u^2_{z_{b'},j}\right)\left\|\boldsymbol{m}_{z_b}-\bar{\boldsymbol{z}}_{z_b}\right\|_2^2}_{T_b^{(3)}}.
\end{aligned}\tag{7}$$

We now solve for the individual terms $T_b^{(1)}$, $T_b^{(2)}$, and $T_b^{(3)}$.

Derivation of $T_b^{(1)}$ First,

$$\begin{aligned}
T_b^{(1)}&=\mathbb{E}\left(1+\sum_{b'=1}^{B}\sum_{j}\delta_{z_b,z_{b'},j}\ u^2_{z_{b'},j}\right)\boldsymbol{u}^\top\boldsymbol{C}_{z_b}^\top\boldsymbol{C}_{z_b}\boldsymbol{u}\\
&=\mathbb{E}\left(1+\sum_{b'=1}^{B}\sum_{j}\delta_{z_b,z_{b'},j}\ u^2_{z_{b'},j}\right)\mathrm{tr}\left(\boldsymbol{u}^\top\boldsymbol{C}_{z_b}^\top\boldsymbol{C}_{z_b}\boldsymbol{u}\right),
\end{aligned}$$

rotating the elements in the trace,

$$= \mathbb{E}\left(1 + \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ u^2_{z_{b'}, j}\right) \mathrm{tr}\left(\boldsymbol{C}_{z_b}^\top \boldsymbol{C}_{z_b} \boldsymbol{u}\boldsymbol{u}^\top\right),$$

grouping the stochastic terms,

$$= \mathrm{tr}\left(\boldsymbol{C}_{z_b}^\top \boldsymbol{C}_{z_b}\ \mathbb{E}\left(1 + \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ u^2_{z_{b'}, j}\right) \boldsymbol{u}\boldsymbol{u}^\top\right)$$

$$= \mathrm{tr}\left(\boldsymbol{C}_{z_b}^\top \boldsymbol{C}_{z_b}\ \left(\mathbb{E}\boldsymbol{u}\boldsymbol{u}^\top + \mathbb{E} \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ u^2_{z_{b'}, j} \boldsymbol{u}\boldsymbol{u}^\top\right)\right)$$

and from Assumption 1,

$$= \mathrm{tr}\left(\boldsymbol{C}_{z_b}^\top \boldsymbol{C}_{z_b}\ \left(\mathbf{I} + \mathbb{E} \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ u^2_{z_{b'}, j} \boldsymbol{u}\boldsymbol{u}^\top\right)\right)$$

$$= \|\boldsymbol{C}_{z_b}\|_{\mathrm{F}}^2 + \mathrm{tr}\left(\boldsymbol{C}_{z_b}^\top \boldsymbol{C}_{z_b}\ \mathbb{E}\left(\sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ u^2_{z_{b'}, j} \boldsymbol{u}\boldsymbol{u}^\top\right)\right).$$

Since $\boldsymbol{C}_{z_b}^\top \boldsymbol{C}_{z_b} \succ 0$ by construction, we can apply Lemma 5 as

$$\leq \|\boldsymbol{C}_{z_b}\|_{\mathrm{F}}^2 + \left\|\mathbb{E} \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ u^2_{z_{b'}, j} \boldsymbol{u}\boldsymbol{u}^\top\right\|_{2,2} \mathrm{tr}\left(\boldsymbol{C}_{z_b}^\top \boldsymbol{C}_{z_b}\right)$$

$$= \|\boldsymbol{C}_{z_b}\|_{\mathrm{F}}^2 + \left\|\mathbb{E} \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ u^2_{z_{b'}, j} \boldsymbol{u}\boldsymbol{u}^\top\right\|_{2,2} \|\boldsymbol{C}_{z_b}\|_{\mathrm{F}}^2$$

$$= \left(1 + \left\|\mathbb{E} \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ u^2_{z_{b'}, j} \boldsymbol{u}\boldsymbol{u}^\top\right\|_{2,2}\right) \|\boldsymbol{C}_{z_b}\|_{\mathrm{F}}^2. \tag{8}$$

Furthermore, from Lemma 7, we have

$$\left\|\mathbb{E} \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ u^2_{z_{b'}, j} \boldsymbol{u}\boldsymbol{u}^\top\right\|_{2,2} = \left(\sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j} + k_\varphi - 1\right) = \left(\sum_{b'=1}^{B} r_{z_b, z_{b'}} + k_\varphi - 1\right).$$

Therefore, back to Equation (8),

$$T_b^{(1)} = \left(1 + \left\|\mathbb{E} \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ u^2_{z_{b'}, j} \boldsymbol{u}\boldsymbol{u}^\top\right\|_{2,2}\right) \|\boldsymbol{C}_{z_b}\|_{\mathrm{F}}^2 = \left(\sum_{b'=1}^{B} r_{z_b, z_{b'}} + k_\varphi\right) \|\boldsymbol{C}_{z_b}\|_{\mathrm{F}}^2. \tag{9}$$

Derivation of $T_b^{(2)}$ Meanwhile, for $T_b^{(2)}$,

$$T_b^{(2)} = \mathbb{E}\left(1 + \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ u^2_{z_{b'}, j}\right) \left(\boldsymbol{m}_{z_b} - \bar{\boldsymbol{z}}_{z_b}\right) \boldsymbol{C}_{z_b} \boldsymbol{u},$$

and from Assumption 1 and Lemma 2,

$$= \left(\boldsymbol{m}_{z_b} - \bar{\boldsymbol{z}}_{z_b}\right) \boldsymbol{C}_{z_b} \mathbb{E}\boldsymbol{u}\ +\ \left(\boldsymbol{m}_{z_b} - \bar{\boldsymbol{z}}_{z_b}\right) \boldsymbol{C}_{z_b} \left(\sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ \mathbb{E} u^2_{z_{b'}, j} \boldsymbol{u}\right)$$

$$= 0. \tag{10}$$

Derivation of $T_b^{(3)}$ Finally,

$$T_b^{(3)} = \mathbb{E}\left(1 + \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ u_{z_{b'}, j}^2\right) \|\boldsymbol{m}_{z_b} - \bar{\boldsymbol{z}}_{z_b}\|_2^2 = \left(1 + \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ \mathbb{E} u_{z_{b'}, j}^2\right) \|\boldsymbol{m}_{z_b} - \bar{\boldsymbol{z}}_{z_b}\|_2^2,$$

and from Assumption 1,

$$\begin{aligned} &= \left(1 + \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\right) \|\boldsymbol{m}_{z_b} - \bar{\boldsymbol{z}}_{z_b}\|_2^2 \\ &= \left(1 + \sum_{b'=1}^{B} r_{z_b, z_{b'}}\right) \|\boldsymbol{m}_{z_b} - \bar{\boldsymbol{z}}_{z_b}\|_2^2. \end{aligned} \tag{11}$$

Combining all the results to Equation (7),

$$\begin{aligned} &\mathbb{E}\left(1 + \sum_{b'=1}^{B} \sum_{j} \delta_{z_b, z_{b'}, j}\ u_{z_{b'}, j}^2\right) \left\|\mathbf{I}_{z_b} \left(\mathcal{T}_\lambda(\boldsymbol{u}) - \bar{\boldsymbol{z}}\right)\right\|_2^2 \\ &\quad \leq T_b^{(1)} - 2T_b^{(2)} + T_b^{(3)}, \end{aligned}$$

applying Equations (9) to (11),

$$\leq \left(\sum_{b'=1}^{B} r_{z_b, z_{b'}} + 1\right) \|\boldsymbol{m}_{z_b} - \bar{\boldsymbol{z}}_{z_b}\|_2^2 + \left(\sum_{b'=1}^{B} r_{z_b, z_{b'}} + k_\varphi\right) \|\boldsymbol{C}_{z_b}\|_{\mathrm{F}}^2.$$

□

B.4.2 Main Result

Theorem 1. *Let Assumption 2 hold. Then, for an L-smooth ℓ, the gradient variance of $\boldsymbol{g}_M(\lambda)$ is bounded as*

$$\mathbb{E}\|\boldsymbol{g}_M(\boldsymbol{\lambda})\|_2^2 \leq \frac{L^2}{M}\left(2\left(d^* + k_\varphi\right) + M\right)\|\boldsymbol{\lambda} - \boldsymbol{\lambda}^*\|_2^2 + \frac{2L^2}{M}\left(d^* + k_\varphi\right)\|\boldsymbol{\lambda}^* - \bar{\boldsymbol{\lambda}}\|_2^2$$

for any $\boldsymbol{\lambda}, \boldsymbol{\lambda}^* \in \Lambda$, where

$$d^* = \max_{b=1,\ldots,B} \sum_{b'=1}^{B} r_{z_b, z_{b'}}$$

is the effective dimensionality, $\bar{\boldsymbol{\lambda}} = (\bar{\boldsymbol{z}}, \mathbf{0})$, and $\bar{\boldsymbol{z}}$ is a stationary point of ℓ.

Proof. The first step mirrors the proof of Theorem 4 by Kim et al. (2023a):

$$\begin{aligned}
\mathbb{E}\|\boldsymbol{g}_M(\boldsymbol{\lambda})\|_2^2 &= \mathbb{E}\left\|\frac{1}{M}\sum_{m=1}^{M}\nabla_\lambda \ell(\mathcal{T}_\lambda(\boldsymbol{u}_m))\right\|_2^2 \\
&= \mathrm{tr}\mathbb{V}\left[\frac{1}{M}\sum_{m=1}^{M}\nabla_\lambda \ell(\mathcal{T}_\lambda(\boldsymbol{u}_m))\right] + \|\nabla f(\boldsymbol{\lambda})\|_2^2 && \text{(Definition of variance)} \\
&= \frac{1}{M^2}\sum_{m=1}^{M}\mathrm{tr}\mathbb{V}\left[\nabla_\lambda \ell(\mathcal{T}_\lambda(\boldsymbol{u}_m))\right] + \|\nabla f(\boldsymbol{\lambda})\|_2^2 && \text{(Linearity of variance)} \\
&= \frac{1}{M}\mathrm{tr}\mathbb{V}\left[\nabla_\lambda \ell(\mathcal{T}_\lambda(\boldsymbol{u}))\right] + \|\nabla f(\boldsymbol{\lambda})\|_2^2 && (\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m \text{ are } i.i.d) \\
&\leq \frac{1}{M}\mathbb{E}\|\nabla_\lambda \ell(\mathcal{T}_\lambda(\boldsymbol{u}))\|_2^2 + \|\nabla f(\boldsymbol{\lambda})\|_2^2, && (12)
\end{aligned}$$

where the last line follows the fact that

$$\mathrm{tr}\mathbb{V}\boldsymbol{x} = \mathbb{E}\|\boldsymbol{x}\|_2^2 - \|\mathbb{E}\boldsymbol{x}\|_2^2 \leq \mathbb{E}\|\boldsymbol{x}\|_2^2$$

for any vector-valued random variable $\boldsymbol{x}$.

For the expected-squared norm term, from (Lemma 8), we now have

$$\mathbb{E}\|\nabla_\lambda \ell(\mathcal{T}_\lambda(\boldsymbol{u}))\|_2^2 = \sum_{b=1}^{B}\mathbb{E}\left(1 + \sum_{b'=1}^{B}\sum_{j}\delta_{z_b, z_{b'}, j}\, u_{z_{b'}, j}^2\right)\left\|\nabla_{z_b}\ell\left(\mathcal{T}_\lambda(\boldsymbol{u})\right)\right\|_2^2.$$

Since $\bar{\boldsymbol{z}}$ is a stationary point, we have $\nabla \ell(\bar{\boldsymbol{z}}) = 0$, and thus

$$= \sum_{b=1}^{B}\mathbb{E}\left(1 + \sum_{b'=1}^{B}\sum_{j}\delta_{z_b, z_{b'}, j}\, u_{z_{b'}, j}^2\right)\left\|\nabla_{z_b}\ell\left(\mathcal{T}_\lambda(\boldsymbol{u})\right) - \nabla_{z_b}\ell\left(\bar{\boldsymbol{z}}\right)\right\|_2^2$$

applying block-wise smoothness through Lemma 4,

$$\leq L^2\sum_{b=1}^{B}\mathbb{E}\left(1 + \sum_{b'=1}^{B}\sum_{j}\delta_{z_b, z_{b'}, j}\, u_{z_{b'}, j}^2\right)\left\|\mathbf{I}_{z_b}\left(\mathcal{T}_\lambda(\boldsymbol{u}) - \bar{\boldsymbol{z}}\right)\right\|_2^2$$

and from Lemma 9,

$$= L^2\sum_{b=1}^{B}\left(\sum_{b'=1}^{B} r_{z_b, z_{b'}} + 1\right)\|\mathbf{I}_{z_b}(\boldsymbol{m} - \bar{\boldsymbol{z}})\|_2^2 + L^2\sum_{b=1}^{B}\left(\sum_{b'=1}^{B} r_{z_b, z_{b'}} + k_\varphi\right)\|\mathbf{I}_{z_b}\boldsymbol{C}\|_{\mathrm{F}}^2.$$

Taking the maximum over the row-wise blocks,

$$\begin{aligned}
&\leq L^2\left(\max_b\sum_{b'=1}^{B} r_{z_b, z_{b'}} + 1\right)\sum_{b=1}^{B}\|\mathbf{I}_{z_b}(\boldsymbol{m} - \bar{\boldsymbol{z}})\|_2^2 + L^2\left(\max_b\sum_{b'=1}^{B} r_{z_b, z_{b'}} + k_\varphi\right)\sum_{b=1}^{B}\|\mathbf{I}_{z_b}\boldsymbol{C}\|_{\mathrm{F}}^2 \\
&= L^2\left(d^* + 1\right)\sum_{b=1}^{B}\|\mathbf{I}_{z_b}(\boldsymbol{m} - \bar{\boldsymbol{z}})\|_2^2 + \left(d^* + k_\varphi\right)\sum_{b=1}^{B}\|\mathbf{I}_{z_b}\boldsymbol{C}\|_{\mathrm{F}}^2
\end{aligned}$$

We can now re-assemble the full vectors/matrices using the properties in Appendix B.1 as

$$= L^2 \left(d^* + 1\right) \|\boldsymbol{m} - \bar{\boldsymbol{z}}\|_2^2 + \left(d^* + k_\varphi\right) \|\boldsymbol{C}\|_{\mathrm{F}}^2. \tag{13}$$

The last steps are standard. We have

$$\mathbb{E}\|\nabla_{\boldsymbol{\lambda}} \ell(\mathcal{T}_{\boldsymbol{\lambda}}(\boldsymbol{u}))\|_2^2 \leq L^2 \left(d^* + 1\right) \|\boldsymbol{m} - \bar{\boldsymbol{z}}\|_2^2 + \left(d^* + k_\varphi\right) \|\boldsymbol{C}\|_{\mathrm{F}}^2,$$

and since the kurtosis satisfies $k_\varphi \geq 1$ by definition,

$$\begin{aligned} &\leq L^2 \left(d^* + k_\varphi\right) \left(\|\boldsymbol{m} - \bar{\boldsymbol{z}}\|_2^2 + \|\boldsymbol{C}\|_{\mathrm{F}}^2\right) \\ &= L^2 \left(d^* + k_\varphi\right) \|\boldsymbol{\lambda} - \bar{\boldsymbol{\lambda}}\|_2^2, \end{aligned} \tag{14}$$

where $\bar{\boldsymbol{\lambda}} = (\bar{\boldsymbol{z}}, \mathbf{0})$ is the variational parameter resulting in q being concentrated on the stationary point of ℓ.

Now, for the gradient norm, Domke (2020, Theorem 1 & Lemma 12) show that when ℓ is L-smooth, the energy f is also L-smooth. Therefore, we can apply smoothness one last time as

$$\|\nabla f(\boldsymbol{\lambda})\|_2^2 = \|\nabla f(\boldsymbol{\lambda}) - \nabla f(\boldsymbol{\lambda}^*)\|_2^2 \leq L^2 \|\boldsymbol{\lambda} - \boldsymbol{\lambda}^*\|_2^2. \tag{15}$$

Combining the results to Equation (12), we have

$$\begin{aligned} \mathbb{E}\|\boldsymbol{g}_M(\boldsymbol{\lambda})\|_2^2 &\leq \frac{1}{M} \mathbb{E}\|\nabla_{\boldsymbol{\lambda}} \ell(\mathcal{T}_{\boldsymbol{\lambda}}(\boldsymbol{u}))\|_2^2 + \|\nabla f(\boldsymbol{\lambda})\|_2^2 \\ &\leq \frac{1}{M} L^2 \left(d^* + k_\varphi\right) \|\boldsymbol{\lambda} - \bar{\boldsymbol{\lambda}}\|_2^2 + \|\nabla f(\boldsymbol{\lambda})\|_2^2 && \text{(Equation (14))} \\ &\leq \frac{1}{M} L^2 \left(d^* + k_\varphi\right) \|\boldsymbol{\lambda} - \bar{\boldsymbol{\lambda}}\|_2^2 + L^2 \|\boldsymbol{\lambda} - \boldsymbol{\lambda}^*\|_2^2 && \text{(Equation (15))} \\ &= \frac{1}{M} L^2 \left(d^* + k_\varphi\right) \|\boldsymbol{\lambda} - \boldsymbol{\lambda}^* + \boldsymbol{\lambda}^* - \bar{\boldsymbol{\lambda}}\|_2^2 + L^2 \|\boldsymbol{\lambda} - \boldsymbol{\lambda}^*\|_2^2 \\ &\leq \frac{1}{M} L^2 \left(d^* + k_\varphi\right) \left(2\|\boldsymbol{\lambda} - \boldsymbol{\lambda}^*\|_2^2 + 2\|\boldsymbol{\lambda}^* - \bar{\boldsymbol{\lambda}}\|_2^2\right) + L^2 \|\boldsymbol{\lambda} - \boldsymbol{\lambda}^*\|_2^2 && \left(|a+b|^2 \leq 2a^2 + 2b^2\right) \\ &= L^2 \left(\frac{2}{M} \left(d^* + k_\varphi\right) + 1\right) \|\boldsymbol{\lambda} - \boldsymbol{\lambda}^*\|_2^2 + L^2 \frac{2}{M} \left(d^* + k_\varphi\right) \|\boldsymbol{\lambda}^* - \bar{\boldsymbol{\lambda}}\|_2^2. \end{aligned}$$

□

B.5 Non-Asymptotic Complexity of Proximal SGD

Lemma 1. *For a μ-strongly convex and L-smooth $f : \Lambda \to \mathbb{R}$ on a convex set Λ and an estimator satisfying Assumption 3, the last iterate $\boldsymbol{\lambda}_{T+1}$ of proximal SGD with a fixed stepsize satisfies $\|\boldsymbol{\lambda}_{T+1} - \boldsymbol{\lambda}^*\|_2^2 \leq \epsilon$ if*

$$\gamma = \min\left(\frac{\epsilon\mu}{4\left(\beta + L^2\bar{\Delta}^2\right)}, \frac{2\alpha}{\mu}, \mu\right) \quad \textit{and}$$

$$T \geq \max\left(\frac{4\left(\beta + L^2\bar{\Delta}^2\right)}{\epsilon\mu^2}, \frac{2\alpha}{\mu^2}, 1\right)\log\left(2\Delta_0^2\,\frac{1}{\epsilon}\right),$$

where $\bar{\boldsymbol{\lambda}}$ is a stationary point of f, $\bar{\Delta} = \|\boldsymbol{\lambda}^* - \bar{\boldsymbol{\lambda}}\|_2$, and $\Delta_0 = \|\boldsymbol{\lambda}^* - \boldsymbol{\lambda}_0\|_2$.

Proof. For an composite objective $F = f + h$, where f is μ-strongly convex and L-smooth, Domke et al. (2023, Thm. 7) show that proximal SGD with a fixed stepsize has a convergence guarantee such that

$$\|\boldsymbol{\lambda}_{T+1} - \boldsymbol{\lambda}^*\|_2^2 \leq (1-\gamma\mu)^T\|\boldsymbol{\lambda}_0 - \boldsymbol{\lambda}^*\|_2^2 + \frac{2\gamma}{\mu}\left(\beta + L^2\|\boldsymbol{\lambda}^* - \bar{\boldsymbol{\lambda}}\|_2^2\right).$$

We can convert this to an iteration complexity guarantee for achieving an ϵ-accurate solution using Lemma A.2 of Garrigos and Gower (2023) where the constants are:

$$\alpha_0 = \|\boldsymbol{\lambda}_0 - \boldsymbol{\lambda}^*\|_2^2, \qquad A = \frac{2}{\mu}\left(\beta + L^2\|\boldsymbol{\lambda}^* - \bar{\boldsymbol{\lambda}}\|_2^2\right), \text{ and } \qquad C = \frac{2\alpha}{\mu}, \mu.$$

That is, we can satisfy $\|\boldsymbol{\lambda}_{T+1} - \boldsymbol{\lambda}^*\|_2^2 \leq \epsilon$ with

$$\gamma = \min\left(\frac{\epsilon\mu}{4\left(\beta + L^2\|\boldsymbol{\lambda}^* - \bar{\boldsymbol{\lambda}}\|_2^2\right)}, \frac{2\alpha}{\mu}, \mu\right) \quad \text{and} \quad T \geq \max\left(\frac{4\left(\beta + L^2\|\boldsymbol{\lambda}^* - \bar{\boldsymbol{\lambda}}\|_2^2\right)}{\epsilon\mu^2}, \frac{2\alpha}{\mu^2}, 1\right)\log\left(2\|\boldsymbol{\lambda}_0 - \boldsymbol{\lambda}^*\|_2^2\,\frac{1}{\epsilon}\right).$$

□

B.6 Non-Asymptotic Complexity of BBVI

Theorem 2. *Let Assumption 2 hold. Then, for a μ-strongly log-concave and L-log-smooth posterior, the last iterate $\boldsymbol{\lambda}_{T+1}$ of BBVI with proximal SGD and an M-sample gradient estimator M satisfies a suboptimality of $\|\boldsymbol{\lambda}_{T+1} - \boldsymbol{\lambda}^*\|_2^2 \leq \epsilon$ if*

$$T \geq 8\frac{\kappa^2}{M}\left(d^* + k_\varphi + \frac{1}{2}M\right)\max\left(\bar{\Delta}^2\frac{1}{\epsilon},\ \frac{1}{2}\right)\log\left(2\Delta_0^2\,\frac{1}{\epsilon}\right)$$

for some fixed stepsize γ, where d^* is the effective dimensionality defined in Theorem 1, $\Delta_0 = \|\boldsymbol{\lambda}_0 - \boldsymbol{\lambda}^*\|_2$ is the distance to the optimum, $\bar{\Delta} = \|\bar{\boldsymbol{\lambda}} - \boldsymbol{\lambda}^*\|_2$ is the distance between the optimum and the stationary point of f, $\kappa = L/\mu$ is the condition number, $\bar{\boldsymbol{\lambda}}$ is the stationary point of the energy f, and $\boldsymbol{\lambda}^*$ is the global optimum.

Proof. It is straightforward that Theorem 1 establishes Assumption 3 for $\boldsymbol{g}_M$ with the constants

$$\alpha = L^2\left(\frac{2}{M}\left(d^* + k_\varphi\right) + 1\right) \quad \text{and} \quad \beta = \frac{2L^2}{M}\left(d^* + k_\varphi\right)\|\boldsymbol{\lambda}^* - \bar{\boldsymbol{\lambda}}\|_2^2 = \frac{2L^2}{M}\left(d^* + k_\varphi\right)\bar{\Delta}^2.$$

Also, from Lemma 1, we know that with a number of iterations at least

$$T \geq \max\left(\frac{4\left(\beta + L^2\bar{\Delta}^2\right)}{\epsilon\mu^2}, \frac{2\alpha}{\mu^2}, 1\right)\log\left(2\|\boldsymbol{\lambda}_0 - \boldsymbol{\lambda}^*\|_2^2\,\frac{1}{\epsilon}\right),$$

we can guarantee an ϵ-accurate solution.

Since our parameterization is linear, a μ-strongly log-concave posterior results in a μ-strongly convex ELBO (Challis and Barber, 2013; Domke, 2020; Kim et al., 2023b). Furthermore, Domke (2020) establishes that a L-log-smooth posterior results in an L-smooth energy f. Thus, we can apply Lemma 1.

Then, the lower bound on the number of iterations can be developed as

$$\begin{aligned}
&\max\left(\frac{4\left(\beta + L^2\bar{\Delta}^2\right)}{\epsilon\mu^2}, \frac{2\alpha}{\mu^2}, 1\right)\log\left(2\|\boldsymbol{\lambda}_0 - \boldsymbol{\lambda}^*\|_2^2\,\frac{1}{\epsilon}\right)\\
&= \max\left(4\left(\frac{2L^2}{M}\left(d^* + k_\varphi\right)\bar{\Delta}^2 + L^2\bar{\Delta}^2\right)\frac{1}{\epsilon\mu^2},\ \ 2L^2\left(\frac{2}{M}\left(d^* + k_\varphi\right) + 1\right)\frac{1}{\mu^2}, 1\right)\log\left(2\Delta_0^2\,\frac{1}{\epsilon}\right)\\
&= \max\left(8\frac{L^2}{\mu^2 M}\bar{\Delta}^2\left(d^* + k_\varphi + \frac{M}{2}\right)\frac{1}{\epsilon},\ \ 4\frac{L^2}{\mu^2 M}\left(d^* + k_\varphi + \frac{M}{2}\right), 1\right)\log\left(2\Delta_0^2\,\frac{1}{\epsilon}\right),
\end{aligned}$$

pulling out the $L^2/\left(\mu^2 M\right)$ factors,

$$= \frac{L^2}{\mu^2}\frac{1}{M}\max\left(8\left(d^* + k_\varphi + \frac{M}{2}\right)\bar{\Delta}^2\frac{1}{\epsilon},\ 4\left(d^* + k_\varphi + \frac{M}{2}\right),\ \frac{\mu^2}{L^2}M\right)\log\left(2\Delta_0^2\,\frac{1}{\epsilon}\right)$$

since the second argument of max is always larger than $2M$, and $\mu^2/L^2 M \leq M$, we have

$$= \frac{L^2}{\mu^2}\frac{1}{M}\max\left(8\left(d^* + k_\varphi + \frac{M}{2}\right)\bar{\Delta}^2\frac{1}{\epsilon},\ 4\left(d^* + k_\varphi + \frac{M}{2}\right)\right)\log\left(2\Delta_0^2\,\frac{1}{\epsilon}\right),$$

pulling out the $8(d^* + k_\varphi + M/2)$ factors,

$$\begin{aligned}
&= \frac{L^2}{\mu^2}\frac{8}{M}\left(d^* + k_\varphi + \frac{M}{2}\right)\max\left(\bar{\Delta}^2\frac{1}{\epsilon},\ \frac{4\left(d^* + k_\varphi + \frac{M}{2}\right)}{8\left(d^* + k_\varphi + \frac{M}{2}\right)}\right)\log\left(2\Delta_0^2\,\frac{1}{\epsilon}\right)\\
&= 8\frac{L^2}{\mu^2}\frac{1}{M}\left(d^* + k_\varphi + \frac{1}{2}M\right)\max\left(\bar{\Delta}^2\frac{1}{\epsilon},\ \frac{1}{2}\right)\log\left(2\Delta_0^2\,\frac{1}{\epsilon}\right)\\
&= 8\frac{\kappa^2}{M}\left(d^* + k_\varphi + \frac{1}{2}M\right)\max\left(\bar{\Delta}^2\frac{1}{\epsilon},\ \frac{1}{2}\right)\log\left(2\Delta_0^2\,\frac{1}{\epsilon}\right).
\end{aligned}$$

□

C PROBABILISTIC MODELS AND DATASETS

C.1 Robust Poisson Regression

We use a robustified version of Poisson regression for modeling count data. This model is known as the Poisson-log-normal model (Cameron and Trivedi, 2013, §4.2.4), which is a "localized" version of regular Poisson regression (Wang and Blei, 2018, §3.2). That is, an additional hierarchy is added to model the local variations of each datapoint. A closely related model is negative binomial regression, which is obtained by setting a conjugate prior to the local noise. Here, the noise is modeled to be log-normal, resulting in a non-conjugate likelihood:

$$\begin{aligned} \eta_i &\sim \mathcal{N}\left(\boldsymbol{x}_i\boldsymbol{\beta} + \alpha, \sigma_\eta\right) \qquad i = 1, \dots, n \\ y_i &\sim \mathsf{Poisson}\left(\exp\left(\eta_i\right)\right) \quad i = 1, \dots, n, \end{aligned}$$

where $(\eta_i)_{i=1}^n$ are the local variables. The global variables are given by the priors

$$\begin{aligned} \sigma_\alpha &\sim \mathsf{Student\text{-}t}_+\left(4, 0, 1\right) \\ \sigma_\beta &\sim \mathsf{Student\text{-}t}_+\left(4, 0, 1\right) \\ \sigma_\eta &\sim \mathsf{Student\text{-}t}_+\left(4, 0, 1\right) \\ \alpha &\sim \mathcal{N}\left(0, \sigma_\alpha\right) \\ \boldsymbol{\beta} &\sim \mathcal{N}\left(\mathbf{0}, \sigma_\beta\right). \end{aligned}$$

We use the rwm5yr German health registry doctor visit dataset (Hilbe, 2011) from the `COUNT` package in R (Hilbe, 2016).

C.2 Item Response Theory

Item response theory (IRT) is a family of models for estimating the response of humans to a set of items, often in the form of exams or questionnaires (Lord et al., 2008). BBVI has recently been shown to be very successful in estimating human ability from large-scale educational examination datasets (Wu et al., 2020). We employ the so-called two-parameter logistic model, or "2PL" model, for which the likelihood is given as

$$\begin{aligned} \mathrm{logit}_i &= \gamma_{\mathrm{item}_i}\alpha_{\mathrm{student}_i} + \beta_{\mathrm{item}_i} + \mu_\beta \\ y_i &\sim \mathsf{Bernoulli\text{-}logit}\left(\mathrm{logit}_i\right) \qquad i = 1, \dots, n. \end{aligned}$$

The global variables are given as

$$\begin{aligned} \mu_\beta &\sim \mathsf{Student\text{-}t}\left(4, 0, 1\right) \\ \sigma_\beta &\sim \mathsf{Student\text{-}t}_+\left(4, 0, 1\right) \\ \sigma_\gamma &\sim \mathsf{Student\text{-}t}_+\left(4, 0, 1\right) \\ \gamma_k &\sim \mathsf{log\text{-}normal}\left(0, \sigma_\gamma\right) \quad k = 1, \dots, K, \end{aligned}$$

while the local variables are

$$\alpha_j \sim \mathcal{N}\left(0, 1\right) \quad j = 1, \dots, J.$$

The log-normal prior on γ is inspired by Patz and Junker (1999).

While here we only consider scaling with respect to the students, we can also consider scaling with respect to the number of items by also making $q\left(\boldsymbol{\beta}\right)$ and $q\left(\boldsymbol{\gamma}\right)$ factor out. While this is less important for our dataset of choice, which has a small K, this is certainly attractive to other datasets where both K and J are large.

For the dataset, we take CritLangAcq from Wu et al. (2020). Unfortunately, this dataset is too large to fit in memory even for the mean-field approximation. Therefore, we only use a 5% random subset of the full dataset. Scaling to the full dataset will require additional strategies amortization (Dayan et al., 1995; Kingma and Welling, 2014) as done in the original work by Wu et al..

C.3 Multivariate Stochastic Volatility

Multivariate stochastic volatility (Chib et al., 2009) The likelihood is given as

$$
\begin{aligned}
\boldsymbol{y}_1 &\sim \mathcal{N}\left(\boldsymbol{\mu}, \boldsymbol{Q}\right) \\
\boldsymbol{y}_t &\sim \mathcal{N}\left(\boldsymbol{\mu} + \boldsymbol{\phi}\left(\boldsymbol{y}_{t-1} - \boldsymbol{\mu}\right), \boldsymbol{Q}\right) \\
\boldsymbol{x}_t &\sim \mathcal{N}\left(0, \exp\left(\boldsymbol{y}_t / 2\right)\right),
\end{aligned}
$$

where $(\boldsymbol{y}_t)_{t=1}^{T}$, the latent stochastic volatilities, are the local variables. For the hyperpriors, we develop a fully Bayesian variant of the model used by Naesseth et al. (2018, §5) who perform empirical Bayes inference on the hyperparameters. Notably, following modern Bayesian modeling practice (Gelman et al., 2020), we assign a Cauchy-LKJ prior to the covariance $\boldsymbol{Q}$. The global variables are given as

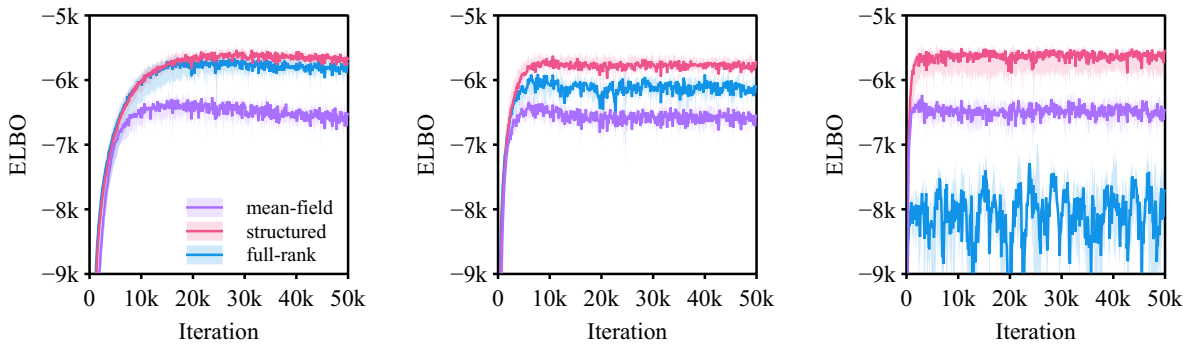
$$
\begin{aligned}
\boldsymbol{L}_{\boldsymbol{\Sigma}} &\sim \mathsf{LKJ\text{-}Cholesky}\left(d_{\boldsymbol{y}}, 1\right) \\
\boldsymbol{\tau} &\sim \mathsf{Cauchy}_{+}\left(0, 5\right) \\
\boldsymbol{L}_{\boldsymbol{Q}} &= \operatorname{diag}\left(\boldsymbol{\tau}\right) \boldsymbol{L}_{\boldsymbol{\Sigma}} \\
\boldsymbol{Q} &= \boldsymbol{L}_{\boldsymbol{Q}} \boldsymbol{L}_{\boldsymbol{Q}}^{\top} \\
\boldsymbol{\mu} &\sim \mathsf{Cauchy}\left(0, 10\right) \\
\boldsymbol{\phi} &\sim \mathsf{uniform}\left(-1, 1\right),
\end{aligned}
$$

where all vector operations are elementwise.

For the datasets, we use the exchange rate ("FX") between 6 international currencies and the U.S. dollar. In particular, we use the daily closing exchange rate of EUR, JPY, GBP, AUD, CAD, and KRW over the period from 2006-05-16 to 2023-08-30. For the subsets, we deterministically slice a continuous period starting from 2006-05-16.

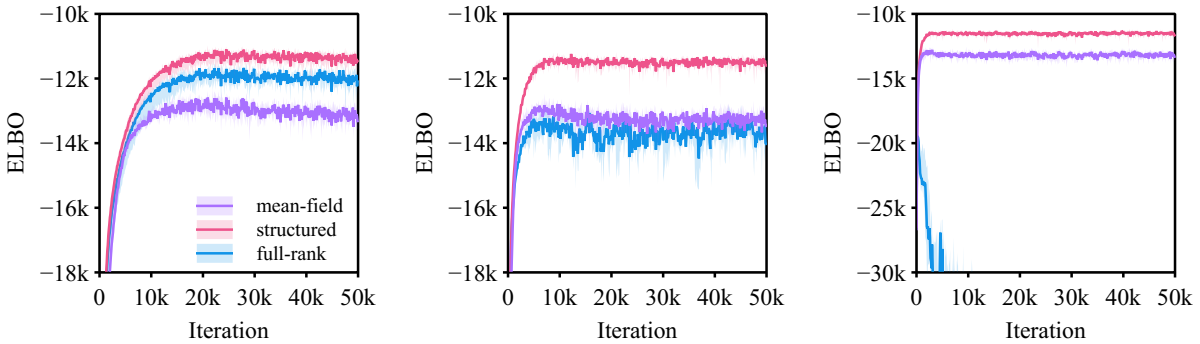
D ADDITIONAL EXPERIMENTAL RESULTS

D.1 Results on rpoisson



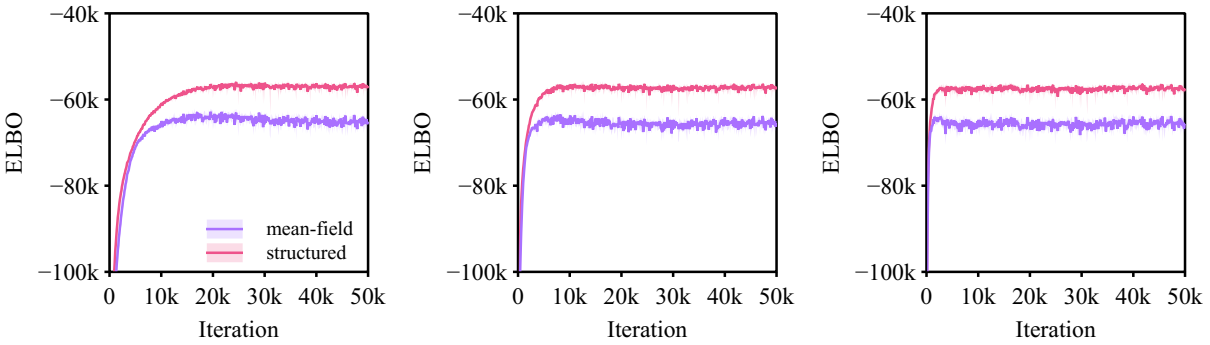
(a) $\gamma = 10^{-4}$ (b) $\gamma = 10^{-3.5}$ (c) $\gamma = 10^{-3}$

Figure 4: **ELBO versus stepsize on rpoisson-small** The solid lines are the median, while the shaded regions are the 80% quantiles computed from 4 independent replications. Notice that the performance gap between **full-rank** and **structured** becomes narrower as we reduce the stepsize.



(a) $\gamma = 10^{-4}$ (b) $\gamma = 10^{-3.5}$ (c) $\gamma = 10^{-3}$

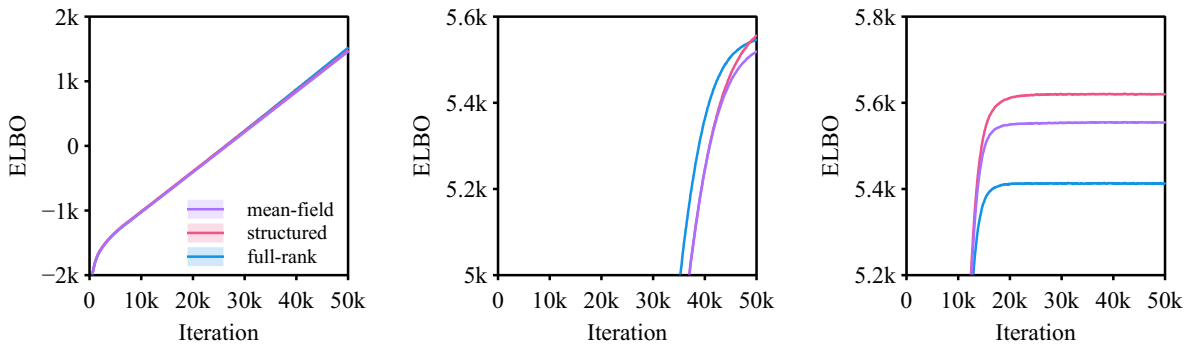
Figure 5: **ELBO versus stepsize on rpoisson-middle** The solid lines are the median, while the shaded regions are the 80% quantiles computed from 4 independent replications. Notice that the performance gap between **full-rank** and **structured** becomes narrower as we reduce the stepsize.



(a) $\gamma = 10^{-4}$ (b) $\gamma = 10^{-3.5}$ (c) $\gamma = 10^{-3}$

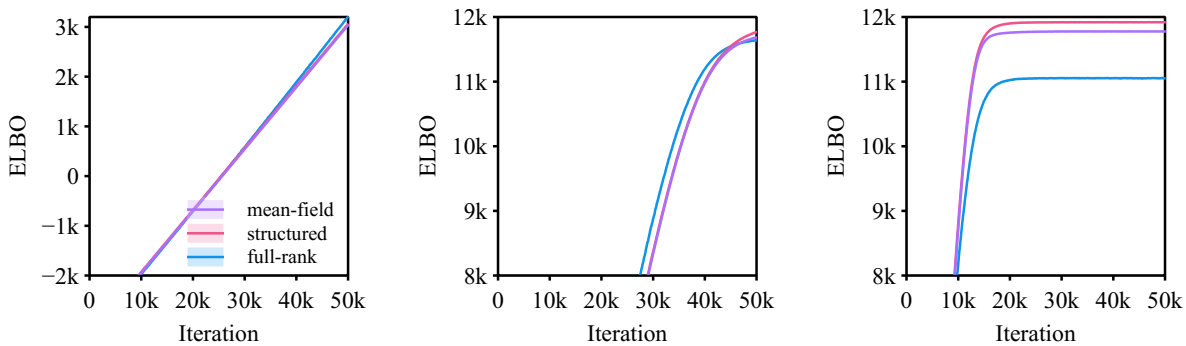
Figure 6: **ELBO versus stepsize on rpoisson-large) Full-rank** is omitted as it didn't fit in memory. The solid lines are the median, while the shaded regions are the 80% quantiles computed from 4 independent replications.

D.2 Results on volatility



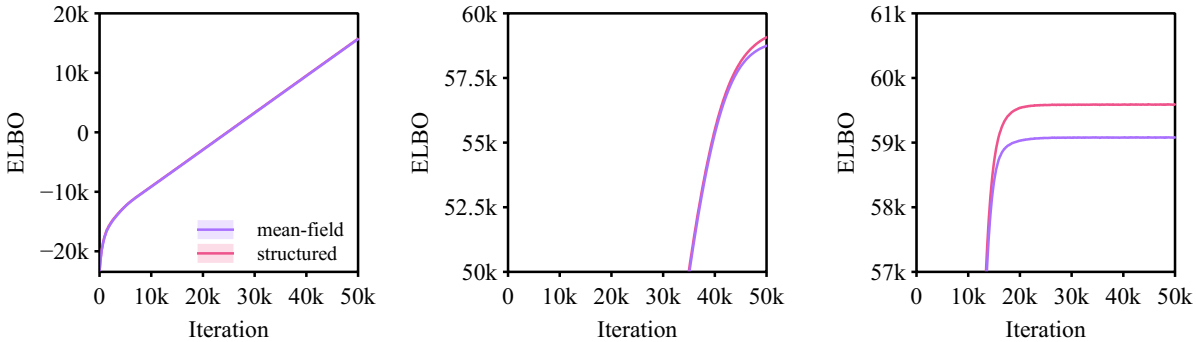
(a) $\gamma = 10^{-4}$ (b) $\gamma = 10^{-3.5}$ (c) $\gamma = 10^{-3}$

Figure 7: **ELBO versus stepsize on volatility-small** The solid lines are the median, while the shaded regions are the 80% quantiles computed from 4 independent replications.



(a) $\gamma = 10^{-4}$ (b) $\gamma = 10^{-3.5}$ (c) $\gamma = 10^{-3}$

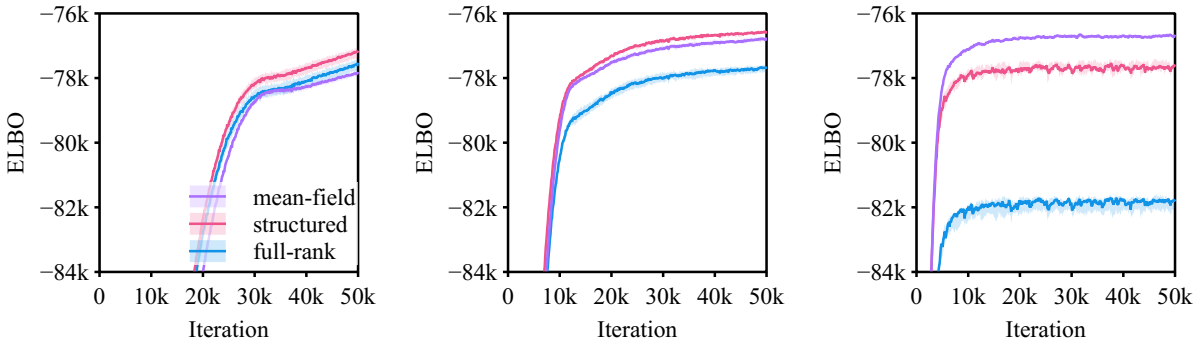
Figure 8: **ELBO versus stepsize on volatility-middle** The solid lines are the median, while the shaded regions are the 80% quantiles computed from 4 independent replications.



(a) $\gamma = 10^{-4}$ (b) $\gamma = 10^{-3.5}$ (c) $\gamma = 10^{-3}$

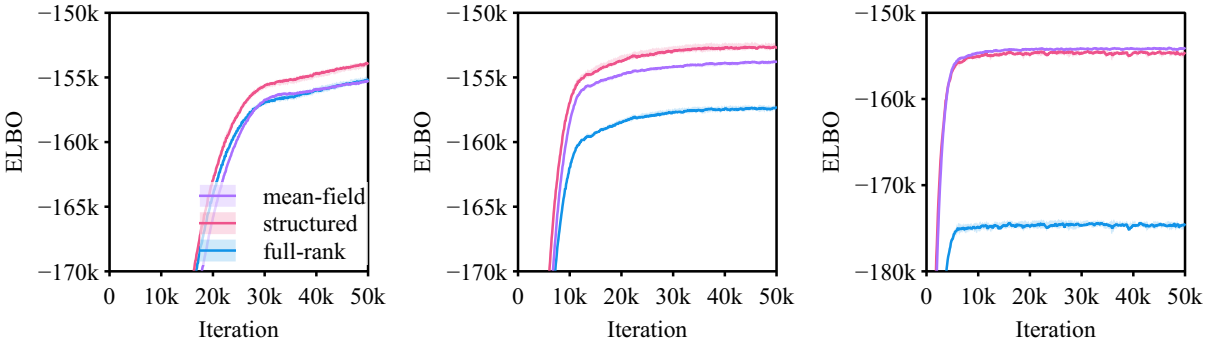
Figure 9: **ELBO versus stepsize on volatility-large Full-rank** is omitted as it didn't fit in memory. The solid lines are the median, while the shaded regions are the 80% quantiles computed from 4 independent replications.

D.3 Results on irt



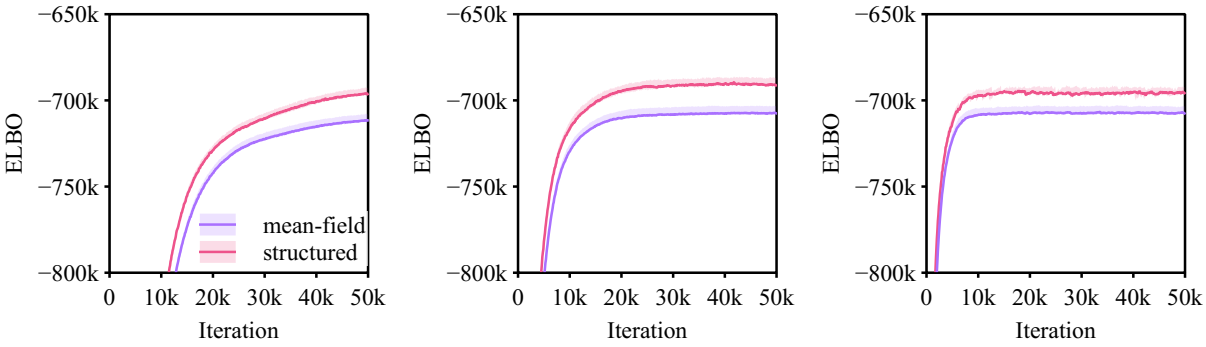
(a) $\gamma = 10^{-4}$ (b) $\gamma = 10^{-3.5}$ (c) $\gamma = 10^{-3}$

Figure 10: **ELBO versus stepsize on irt-small** The solid lines are the median, while the shaded regions are the 80% quantiles computed from 4 independent replications. Notice that the **structured** performs worse than mean-field at the largest stepsize ($\gamma = 10^{-3}$), but becomes superior as we reduce the stepsize.



(a) $\gamma = 10^{-4}$ (b) $\gamma = 10^{-3.5}$ (c) $\gamma = 10^{-3}$

Figure 11: **ELBO versus stepsize on irt-middle** The solid lines are the median, while the shaded regions are the 80% quantiles computed from 4 independent replications. Notice that the **structured** performs slightly worse than **mean-field** at the largest stepsize ($\gamma = 10^{-3}$), but becomes superior as we reduce the stepsize.



(a) $\gamma = 10^{-4}$ (b) $\gamma = 10^{-3.5}$ (c) $\gamma = 10^{-3}$

Figure 12: **ELBO versus stepsize on irt-large Full-rank** is omitted as it didn't fit in memory. The solid lines are the median, while the shaded regions are the 80% quantiles computed from 4 independent replications.